Online Appendix for "RefSel: Risk-Controlled Foreign-Key Discovery with Bounded Language-Model Evidence"

Jiamao Lin and Weisong Sun

This document accompanies the main paper. Section, table, figure, equation, and algorithm numbers with the prefix A refer to this document; all other numbers refer to the main paper. Citations refer to the reference list at the end of this document.

A PROOFS

Proof of Lemma 1. (a) Under H_0, all $\binom{u}{k}$ size-k subsets of $\mathcal{U}$ are equally likely, and exactly $\binom{N}{k}$ of them lie entirely inside P. Because $N \le u$, each factor in the product $\binom{N}{k}/\binom{u}{k} = \prod_{i=0}^{k-1} \frac{N-i}{u-i}$ satisfies $\frac{N-i}{u-i} \le \frac{N}{u} = q$, yielding $\Pr_{H_0}[X = k] \le q^k$ and thus $s_e = -\log \Pr_{H_0}[X = k] \ge k \log(u/N)$. (b) By Hoeffding's theorem [1, Thm. 4], hypergeometric sampling without replacement is dominated in the convex order by binomial sampling with replacement $\mathrm{Bin}(k, q)$; consequently, the standard Chernoff–Sanov upper tail bound $\Pr[\mathrm{Bin}(k, q) \ge \theta k] \le \exp(-k\,\mathrm{KL}(\theta \| q))$ for $\theta > q$ applies directly to $X \sim \mathrm{Hyp}(u, N, k)$. Note that setting $\theta = 1$ recovers statement (a) since $\mathrm{KL}(1\|q) = \log(1/q)$. (c) Because $e^{-s_e} = \Pr_{H_0}[X \ge m]$ is the upper-tail p-value of a discrete test statistic under H_0, it is stochastically larger than a standard uniform random variable on $[0, 1]$: for every $t \ge 0$, $\Pr_{H_0}[s_e \ge t] = \Pr_{H_0}[e^{-s_e} \le e^{-t}] \le e^{-t}$. Integrating this tail bound over $t \in [0, \infty)$ yields $\mathbb{E}_{H_0}[s_e] = \int_0^\infty \Pr_{H_0}[s_e \ge t]\, dt \le \int_0^\infty e^{-t}\, dt = 1$. □

In our implementation, the hypergeometric tail in Eq. (3) is evaluated in log-space using log-gamma functions over at most 4,000 terms (whose probabilities decay geometrically past the mode) for domain sizes up to $u \le e^{25}$, and via the exact binomial tail approximation for $u > e^{25}$; whenever the observed overlap m falls at or below the mode of $\mathrm{Hyp}(u, N, k)$, s_e is set to the closed-form lower bound of Lemma 1(b).

Nested targets and transitive shortcuts. Because the binomial coefficient $\binom{N}{i}$ is strictly increasing in N, a smaller target key P_1 that still contains S renders the observed inclusion strictly more surprising under H_0 than a larger superset key P_2.

Lemma A1 (Tightest containment)**.** *Suppose a source set S satisfies $S \subseteq P_1 \subseteq P_2$ for two candidate targets $e_1, e_2 \in g$, and suppose both candidates are evaluated against a common domain size u. Then $s_{e_1} \ge s_{e_2}$, with equality if and only if $|P_1| = |P_2|$.*

Proof. Because $S \subseteq P_1 \subseteq P_2$, both candidates have exact coverage $m = k = |S|$. By Lemma 1(a), $s_e = \log\binom{u}{k} - \log\binom{N}{k}$, which is strictly decreasing in $N = |P|$ for all $N \ge k$, and $N_1 = |P_1| \ge k$ holds because $S \subseteq P_1$. □

Because set inclusion is transitive ($A.X \subseteq B.Y$ and $B.Y \subseteq C.Z$ automatically imply $A.X \subseteq C.Z$), any multi-hop reference chain—such as a subtype hierarchy *Employee* $\subseteq$ *Person* $\subseteq$ *BusinessEntity* in AdventureWorks—generates transitive shortcut candidates pointing from the leaf table directly to the root table. Evaluated under a shared domain u, Lemma A1 guarantees that the true, immediate parent target P_1 is at least as significant as the coarser root target P_2 (though per-candidate domain estimators can occasionally invert this ordering). Crucially, Lemma A1 requires true nesting ($P_1 \subseteq P_2$): if P_1 is merely a tiny unrelated lookup table that happens to cover the values of a low-cardinality column S without being a subset of the true target P_2, preferring the smaller target P_1 produces a false positive. We evaluated an explicit Stage 3 decoder potential that rewards the tightest nested target; as reported in Section 5.4 and Table A4, it did not improve end-to-end F1.

Proof of Proposition 1. Fix any permutation σ of $\{1, \ldots, \nu\}$ and let ρ denote the table-and-column relabeling that leaves the anonymous observation O invariant ($\rho(O) = O$) while mapping each candidate e_i to $e_{\sigma(i)}$. By relabeling invariance of the joint law of (O, R), we have $\Pr[O = o, R = r] = \Pr[O = \rho(o), R = \rho(r)] = \Pr[O = o, R = \rho(r)]$ since $\rho(o) = o$. Conditioning on $O = o$ therefore yields $\Pr[R = e_i \mid O] = \Pr[R = e_{\sigma(i)} \mid O]$. Varying σ over all permutations implies that all ν conditional target probabilities are identical, and since they sum to $\pi_g = \Pr[R \ne \perp \mid O]$, we obtain $\Pr[R = e_i \mid O] = \pi_g/\nu$ for every $i \in \{1, \ldots, \nu\}$. Finally, any decision rule whose output $\hat{e}$ depends only on O and independent internal randomness satisfies $\Pr[\hat{e} = R \mid O, R \ne \perp] = \sum_{i=1}^{\nu} \Pr[\hat{e} = e_i \mid O] \cdot \frac{1}{\nu} \le \frac{1}{\nu}$. □

While exact automorphism of an entire database instance O is rare, the exact same $1/\nu$ limit holds at the level of feature representations, which directly governs learned classifiers such as Stage 2 of RefSel.

Proposition A1 (Feature-level indistinguishability)**.** *Suppose a discovery method observes the database strictly through a feature*

map ϕ, and suppose $\phi(e_1) = \cdots = \phi(e_\nu)$ for ν competing candidates inside source group g. If the conditional distribution of R given ϕ is invariant under permutations of the candidate labels $\{1, \ldots, \nu\}$, then the calibrated posterior probability of each candidate is $\Pr[R = e_i \mid \phi] = \Pr[R \neq \perp \mid \phi]/\nu$, and the method's conditional target-selection accuracy on group g given $R \neq \perp$ is at most $1/\nu$.

Proof. The proof of Proposition 1 applies verbatim upon replacing the raw observation O with the feature representation ϕ. □

This invariance condition holds naturally for local features computed purely from key value sets ($\boldsymbol{\xi}_e, \boldsymbol{\gamma}_e$), whereas schema-wide stacked relational features ($\boldsymbol{\alpha}_e$, Section 3.3) can break feature ties whenever the surrounding table graph is asymmetric.

B IMPLEMENTATION DETAILS

Adaptive Stage 1 profiling budgets. Rather than imposing a rigid global cutoff across all schemas, Stage 1 scales its search budgets with database size: our default profile allocates up to 3,600 s per database, at most 40 composite column combinations per source group, 4,000 composite semi-join verification queries per database, and 300 column pairs plus 300 triples per table for minimal UCC discovery. Large schemas (exceeding 2×10^6 total rows, 100 tables, or 1,500 columns) receive double the wall-clock and combination budgets, whereas compact schemas (at most 2×10^5 rows, 400 total columns, and 40 columns per table) expand composite search up to arity four. Because candidate enumeration is ordered by descending unary coverage, budget truncation—when triggered on ultra-wide schemas—discards the lowest-coverage candidates first. Zero declared foreign keys were lost to budget truncation across all internal and external benchmarks; budget truncation and deadline truncation are distinct: in each Round 3 naming condition, 42 of 89 calibration schemas and 30 of 60 test schemas have some budget or deadline truncation; 14 WikiDBs schemas hit the wall-clock deadline in at least one naming condition. The same frozen partial candidate sets are shared by every method (Section 5.6). On the seven classic external benchmarks, Stage 1 candidate recall is 1.000 under the evaluable-key denominator; under the whole-database denominator, candidate recall on the 370-table MusicBrainz benchmark is 0.581 solely because the remaining declared foreign keys reside in empty source tables or all-null/constant source columns (Table A5).

Same-origin deduplication rules. To guarantee zero schema leakage across development, calibration, and test splits, two databases are classified as sharing the same origin if: (i) their normalized names belong to the same stem family after stripping version and split suffixes, (ii) at least two tables with matching names cover $\geq 80\%$ of the tables in the smaller schema, or (iii) their row-level value fingerprints match. No same-origin pair ever crosses a fold or repository boundary.

Stage 2 feature inventory. The local *edge features* $\boldsymbol{\xi}_e$ (Section 3.3) comprise: distinct coverage θ_e and dangling fraction $1 - \theta_e$; conservative inclusion significance s_e^{cons}, the four individual domain-specific significance estimates, and the analytical KL bound $k\,\mathrm{KL}(\theta\|q)$; distinct cardinality ratio k_e/N_e; source uniqueness and null fraction; atomic type-compatibility indicators; string-length and numeric-range summary moments; and—exclusively in the original naming condition—lexical string similarity between source and target table/column identifiers. The *group competition features* $\boldsymbol{\gamma}_e$ comprise: group candidate count $|g(e)|$; relative rank of e within $g(e)$ by coverage θ_e and by target cardinality N_e; log-significance gap relative to the group leader; count of candidates in $g(e)$ exceeding strong significance ($s_f > \log 10^3$); group mean, minimum, and maximum of every significance feature; and the value-equivalence class size ν_e.

Domain size estimators. Because the true generative domain $\mathcal{U}$ is unobserved, we compute four domain-size estimates for Eq. (3), clipping each from below by $|S \cup P| = N + k - m$:

- *Global range (u^{rng}).* For integer, scaled-decimal, date, and timestamp columns, $u^{\text{rng}} = \max(S \cup P) - \min(S \cup P) + 1$ discrete lattice points.
- *Local source-interior range (u^{loc}).* For ordered numeric and temporal types, we also condition on the observed endpoints $\min S$ and $\max S$ of the source column. Under H_0, the remaining $k' = k - n_{\text{end}}$ interior source values are drawn uniformly without replacement from the u' discrete lattice points in the open interval $(\min S, \max S)$. We evaluate Eq. (3) using u' (clipped below by $N' + k' - m'$), interior target count $N' = |P \cap (\min S, \max S)|$, and interior match count $m' = m - m_{\text{end}}$, where $n_{\text{end}} \in \{1, 2\}$ counts distinct source endpoints and m_{end} counts matched endpoints. This local estimator is critical for exposing prefix collisions: whenever a target key $P = \{1, \ldots, N\}$ densely covers the source interval $(\min S, \max S)$, we have $N' = u'$ and thus $s_e^{\text{loc}} = 0$.
- *Database-wide type cardinality (u^{typ}).* u^{typ} counts the total number of distinct values observed across all columns belonging to the same atomic type family anywhere in database D (estimated via HyperLogLog sketches and taken as the maximum of the source and target type families).
- *Character pattern capacity (u^{pat}).* For string columns, $u^{\text{pat}} = \sum_{L=L_{\min}}^{L_{\max}} |\Sigma|^L$, where Σ is the union of characters observed in $S \cup P$ and $[L_{\min}, L_{\max}]$ is their observed length span; for standard UUID columns, $u^{\text{pat}} = 2^{122}$.

For composite candidates ($|X| \geq 2$), we treat component dimensions as independent and sum their log domain sizes $\log u$. Stage 2 receives the conservative minimum s_e^{cons} (Eq. (4)), the individual estimates, and the KL lower bound of Lemma 1(b) evaluated at $\min(u^{\text{rng}}, u^{\text{typ}}, u^{\text{pat}})$.

Stage 3 integer linear programming solver. All objective coefficients in Eq. (5) are scaled by 10^3 and rounded once to exact integers, and preprocessing bounds use identical integer coefficients so that pruning never alters the global optimum. Solving Eq. (5) with OR-Tools CP-SAT [2] under a 20 s wall-clock cap proves global optimality on all 284 internal named databases and 354 of 355 anonymous databases (a single wide ReDeLEx schema hits the 20 s cap and returns its best feasible solution), with a median solve time of ~ 1 ms and a 95th percentile below 10 ms.

Computational complexity. Stage 1 profiling and semi-join verification scale with the number of type-compatible

column pairs bounded by the adaptive budget. Evaluating hypergeometric surprisal s_e takes $O(\min(k, N) - m + 1)$ log-gamma evaluations per candidate, and computing Stage 2 stacked features is strictly linear in $|\mathcal{E}(D)|$. The Stage 3 ILP contains $O(|\mathcal{E}(D)| + \sum_T |\mathcal{K}_T|)$ binary variables and linear constraints, solving in milliseconds on standard schemas (on massive ERP schemas with 10^5–10^6 candidates, CP-SAT returns high-quality feasible solutions within the 20 s cap; Section 5.7). Stage 4 and 5 confidence and retention require $O(|\hat{F}(D)|)$ time.

C Risk calibration

Learn-then-Test fixed-sequence calibration. Before observing calibration labels, we fix a descending grid of M candidate confidence thresholds $\tau_1 > \tau_2 > \cdots > \tau_M$ chosen from empirical quantiles of out-of-fold development confidences. Given n independent calibration risk units with empirical mean loss $\hat{R}_n(\tau) = n^{-1} \sum_{j=1}^{n} L_j(\tau)$ at threshold τ, we compute the finite-sample valid Hoeffding–Bentkus upper-tail p-value [3] for the unsafe null hypothesis $H_\tau : R(\tau) > \alpha$:

$$p^{\mathrm{HB}}(\tau) = \min\Big\{ \exp\big(-n\, h_1(\hat{R}_n(\tau) \wedge \alpha, \alpha)\big), \\ e \cdot \Pr\big[\mathrm{Bin}(n, \alpha) \le \lceil n \hat{R}_n(\tau) \rceil\big] \Big\}, \tag{A1}$$

where $h_1(a, b) = a \log \frac{a}{b} + (1 - a) \log \frac{1-a}{1-b}$ is the binary relative entropy and $a \wedge \alpha = \min(a, \alpha)$. Because the threshold grid is ordered from strictest (τ_1) to loosest (τ_M), fixed-sequence testing [4] controls family-wise error without Bonferroni dilution by selecting the lowest threshold up to which every stricter grid point rejects H_{τ_i} at level δ:

$$\hat{\tau} = \tau_{j^*}, \qquad j^* = \max\{\, j : \; p^{\mathrm{HB}}(\tau_i) \le \delta \;\; \forall i \le j \,\}. \tag{A2}$$

If even the strictest threshold fails ($p^{\mathrm{HB}}(\tau_1) > \delta$), Stage 5 abstains completely ($\hat{F}_\alpha(D) = \emptyset$).

Exact vs. approximate assumptions in experiments. The Hoeffding–Bentkus inequality (Eq. (A1)) requires independent risk units with losses bounded in $[0, 1]$, matching the i.i.d. setting of Learn-then-Test [4]. Because candidate edges within a single database—and multiple variants of the same database schema—are statistically dependent, we define risk units U_j as intact duplicate-free schema families rather than individual databases or individual edges [5]. As noted in Section 3.5, Proposition 2 bounds the population expectation $R(\hat{\tau}) = \mathbb{E}[L_{\mathrm{new}}(\hat{\tau})]$ across new risk units rather than bounding every individual schema's FDP with high probability. In our five-fold internal cross-validation (Section 5.5), within each outer training split (4 inner folds) we fit the isotonic calibrator $\hat{\pi}_g$ on inner folds 0–1 and calibrate $\hat{\tau}$ on inner folds 2–3 ($n = 107$ to 123 calibration units at $\delta = 0.10$). Two aspects of this cross-validated setup approximate Proposition 2: (i) the scalar acceptance bias b is tuned across all four inner folds, and (ii) inner calibration uses out-of-fold predictions from 3-fold inner models whereas outer test folds are scored by the model refit on all 4 inner folds. Our three independent rounds (Sections 5.6–5.6) eliminate both approximations by locking a single frozen program before touching dedicated out-of-repository calibration and test units.

D Audit of confident errors on the independent data

To understand why LTT threshold calibration stopped early on the $n = 24$ CTU calibration set in Round 1 (Section 5.6), we manually audited every decoded edge in Round 1 that was not listed in the declared SQL ground truth G_D and either (i) scored above the first failing LTT grid threshold at $\alpha = 0.20$ or (ii) ranked among the top 25 false positives per line and naming condition—totaling 74 audited edges on the CTU calibration part and 26 on the RelBench/4DBInfer test part. For each edge, read-only SQL queries extracted original column names and atomic types, distinct cardinalities, coverage θ_e, target uniqueness, whether the same source columns participate in another declared foreign key, and sample rows. Each edge was assigned a conservative audit category: *undeclared reference* was assigned only when column names, domain semantics, and target uniqueness jointly confirmed an intended foreign key omitted by the schema author.

Across the 74 audited calibration edges, 9 are genuine undeclared foreign keys, 3 are value-equivalent rival targets, 2 are ambiguous, and 60 are true errors; across the 26 audited test edges, 24 are genuine coincidental inclusions between dense surrogate counters and 0 are undeclared references. Focusing on the 10 pivotal calibration edges that halted the LTT threshold walk at $\alpha = 0.20$: six are genuine undeclared references in CTU schemas (e.g., `teams.franchID` → `teamsfranchises` in Lahman baseball and `train.(Date,Store)` → `features` in Walmart), one is a value-equivalent rival target, and three are genuine anonymous-regime errors (`indis.in_id` → `rel12` in Hepatitis reverses a 1-to-1 key, a binary $0/1$ column in `geneea` matches a 7-row lookup table, and `invoices.number` → `voyages.artificial_id` in `voc` overlaps two surrogate counters). When the six undeclared CTU references are counted as true positives in a post hoc sensitivity check, the next stopping errors are genuine (kinship schemas `Dunur`, `Elti`, and `Mooney_Family` forming one source group, plus small integer count columns), relaxing the named thresholds (e.g., GBDT $\sigma(t_e) \ge 0.9977$) so that retained true foreign keys per database at $\alpha = 0.20$ rise to 7.8 (GBDT, σ), 6.2 (RefSel, σ), and 6.4 (RefSel, c_e); in the anonymous condition, the three genuine errors still keep RefSel's thresholds unchanged. Full audit logs and SQL verification evidence are included in our artifact repository.

E Use cases, algorithm, and supplementary results

Table A1 maps the operational scenarios discussed in Section G to the benchmark datasets and experimental sections that evaluate them, while Algorithm A1 provides the complete step-by-step training and inference pseudocode aligned with Stages 1–5 of Fig. 1.

Downstream Text-to-SQL experiment details: We evaluated DeepSeek V4.1 Flash across all 1,070 BIRD development queries whose gold SQL contains at least one join and executes cleanly on DuckDB, comparing four schema-prompt conditions: ground-truth declared foreign keys (0.550 execution accuracy), no foreign keys (0.539), all foreign keys

TABLE A1
Use cases, the data that represent them, and where they are evaluated.

Use case	Why foreign keys are missing	Data in this paper	Where
LLM agents and text-to-SQL	warehouses do not enforce keys; exports carry none	BIRD, Spider; downstream text-to-SQL on BIRD	RQ1, RQ4
ERP implementation and migration	integrity kept in application code; partial declarations	Dolibarr (440 tables), Apache OFBiz (857), iDempiere (912)	RQ4
SAP data integration	relationships defined in the application layer	SAP SALT sales data (4 tables, 4.7M rows)	RQ4
Data catalogs and governance	tools read declared constraints only	risk-controlled retention	RQ3
De-identified data sharing	names masked or encoded	anonymous condition throughout	RQ1–RQ4
Data lakes and file exports	files carry no constraints	full MusicBrainz dump; TPC-H and TPC-DS at scale factor 10	RQ4
BI and semantic layers	join graphs built by hand	TPC-DS, AdventureWorks DW; Auto-BI-style decoding	RQ1, RQ4

Algorithm A1 RefSel: training and inference across Stages 1–5

Require: training databases with declared keys; test database D; target risk level (α, δ)

1: **Training.** *(Stage 1)* Generate $\mathcal{E}(D')$ and minimal UCC candidates $\mathcal{K}_T$ for every training database D' (Eq. (1)).
2: *(Stage 2)* Compute local and group features $\boldsymbol{\xi}_e, \boldsymbol{\gamma}_e$ including conservative inclusion significance s_e^{cons} (Eqs. (3)–(4)) and equivalence class size ν_e.
3: Fit first-pass GBDT f_1 via group-disjoint cross-fitting; compute out-of-fold logits $z_e^{(1)}$ and stacked relational features $\boldsymbol{\alpha}_e$ (Eqs. (A3)–(A6)); fit second-pass GBDT f_2 (Eq. (A7)).
4: *(Stage 3)* Select acceptance bias b on inner validation folds.
5: *(Bounded LLM)* Under original names, fix (μ, B) for Eqs. (8)–(9) and verifier confidence multipliers.
6: *(Stages 4–5)* Fit isotonic group calibrator ψ, fix the threshold grid, and run LTT calibration (Eqs. (A1)–(A2)) on calibration risk units to obtain $\hat{\tau}$.
7: **Inference on** D**.** *(Stages 1–2)* Generate $\mathcal{E}(D)$ and features; compute second-pass logits $z_e = f_2(\boldsymbol{\xi}_e, \boldsymbol{\gamma}_e, \boldsymbol{\alpha}_e)$ and shifted scores $t_e = z_e + b$.
8: *(Bounded LLM prior)* Under original names, query the LLM on near-tied groups $\mathcal{Q}(D)$ and form t_e^{L} via Eq. (9).
9: *(Stage 3 & LLM verifier)* Solve the joint ILP (Eq. (5)) over $\mathcal{C}(D)$ exactly; under original names, run the batch verifier on decoded edges, veto edges judged `no`, and weight confidences.
10: *(Stages 4–5)* **return** decoded keys and foreign keys $\hat{F}(D)$ with confidences c_e (Eq. (6)), and the risk-controlled subset $\hat{F}_\alpha(D) = \{e \in \hat{F}(D) : c_e \geq \hat{\tau}\}$.

decoded by RefSel (0.545), and only the subset retained by RefSel at $\alpha = 0.20$ (0.546). While all overall paired differences have 95% confidence intervals that include zero (declared minus none: $+0.011$, CI $[-0.003, 0.025]$), focusing on the 443 hardest join queries where joined columns have dissimilar names shows that ground-truth keys add $+0.018$ accuracy and RefSel keys add $+0.016$, reducing queries with missing gold joins from 209 (no keys) down to 193 (declared) and 200 (retained). Because BIRD schemas contain only 3–13 well-named tables, LLMs already guess most joins from column names alone; measuring downstream utility on wide, cryptically named enterprise schemas remains an important direction for future work.

Ground-truth incompleteness across sibling schemas: Because real schema authors occasionally omit foreign-key declarations, a discovery method that recovers an undeclared true reference is penalized for a false positive under standard benchmarks. To quantify whether such ground-truth omissions distort our internal comparisons, we exploited the 36 duplicate-schema families (covering 84 databases) inside BIRD and Spider: any predicted foreign key that is undeclared in database D but explicitly declared on the matching columns of a sibling copy D' is credited as correct (while keeping recall denominators unchanged). Among foreign keys declared in one sibling copy, 70% map to existing columns in the companion copy, and sibling declarations agree on 97% of those mapped pairs (only 24 of 833 differ across 6 of 34 multi-copy families). Applying this sibling-augmented ground truth increases pooled precision for every evaluated method by at most $+0.005$ across all 279 BIRD/Spider databases (and by at most $+0.030$ on the 42 schemas that possess a sibling copy), confirming that relative rankings in Table 1 are insensitive to inter-copy declaration omissions.

Learned structural graph potentials: The structural potential ablation in Table A4 augments the Stage 3 linear objective (Eq. (5)) with a higher-order structural score $\mathbf{w}^\top \boldsymbol{\Psi}(x, y)$ comprising: (i) a nested-tightness indicator rewarding the smallest nested target per Lemma A1, (ii) a triangle-shortcut penalty discouraging $A \to C$ when $A \to B \to C$ is selected, and (iii) table-level in-degree and participation potentials, with weight vector $\mathbf{w}$ trained via an averaged structured perceptron using loss-augmented CP-SAT inference [6], [7].

Runtime and API cost details: Stage 1 candidate generation takes $\sim$ 19 minutes on Dolibarr (440 tables, 5,755 columns) and reaches our 2-hour cap on iDempiere with original names (in addition to the scaling points shown in Fig. 11). At roughly $1 USD in total API cost across all seven external benchmarks, our group-batched LLM prior requires an order of magnitude fewer calls than per-candidate agentic

prompting on large schemas (Fig. 9).

F Supplementary Experimental Details and Tables

This appendix provides the detailed experimental protocols and complete breakdown tables supporting Section 5: the internal benchmark composition and candidate recall (Table A2), cross-partition stability across three random fold splits (Table A3), the full architectural ablation study (Table A4), per-database accuracy on the seven external benchmarks (Table A5), per-database LLM prior call counts (Table A6), backbone LLM sensitivity across DeepSeek V4.1 Flash and V4 Pro (Table A7), one-time results on the ten pre-registered databases (Table A8), and one-time results on the 14-database CTU confirmation set (Table A9).

Hyperparameter tuning budgets: Every tunable parameter is selected strictly on inner validation folds of the training split: the baseline GBDT classifier searches over $3 \times 25 = 75$ combinations of tree leaf count and acceptance bias b; RefSel tunes only its scalar acceptance bias b (over a 0.5-step grid); Fast-FK searches over 180 heuristic feature-weight configurations; and MC-FK tunes its distribution distance cutoff over 18 grid values (whereas Rostin-style, HoPF-style, and the inclusion rule use fixed published parameters, and Auto-BI-lite reuses the tuned GBDT scores). Both passes (f_1, f_2) of the RefSel scorer use a fixed learning rate of 0.05 and 300 boosting trees: f_1 fixes 31 leaves, and f_2 inherits the leaf count selected for the baseline GBDT classifier in the same outer fold—ensuring that the baseline GBDT classifier receives the strictly larger hyperparameter tuning budget.

Fold partitions and anonymization stability: To quantify sensitivity to the five-fold cross-validation split, we repeated the internal evaluation across two additional random group-disjoint partitions (P1 and P2 in Table A3; slow MC-FK and LLM baselines were not re-executed), computing paired confidence intervals over per-database F1 averaged across P0, P1, and P2. For schema anonymization, our main tables use a single seeded random permutation of table/column names and physical column positions per database; re-anonymizing ten schemas with a fresh random seed alters only physical-position tie-breaking features, changing decoded edges on two schemas and shifting F1 on just one.

External benchmark dataset versions: Our local instances of two external benchmarks differ slightly from those used in LLM-FK [8]: our MusicBrainz relational sample contains 370 tables and 762 declared foreign keys (versus 382 tables and 354 foreign keys in LLM-FK), and our TPC-E instance—generated via the official TPC EGen v1.14.0 data generator [9] and DBT-5 PostgreSQL schema [10]—contains 33 tables and 49 declared foreign keys.

G Motivating Example and Use Cases

Fig. A1 shows the two inclusion traps of the Sakila database discussed in Section 1 and the decisions of RefSel and three baselines on them.

Every experimental condition we study mirrors a concrete operational scenario in which foreign keys or informative column names are absent in practice. First, modern web application frameworks routinely enforce referential integrity in application code rather than in the database engine [16]. Second, packaged enterprise resource planning (ERP) systems define hundreds of logical relationships in their application dictionary while declaring only a fraction as physical SQL constraints, as our evaluation on production open-source ERP schemas confirms. Third, cloud data catalogs record primary and foreign keys only as unenforced informational metadata [17], and data exports to parquet lakes or CSV archives strip schema constraints altogether. Column and table names degrade for equally concrete operational reasons that correspond directly to the naming degradation levels evaluated in Section 5.3: vendor-specific table prefixes such as `llx_` in Dolibarr or `C_` in iDempiere (*prefix* level), cryptic domain abbreviations of the kind pervasive in legacy SAP tables (*abbreviation* level), and auto-generated or privacy-masked identifiers produced by low-code platforms, ETL pipelines, and de-identified data-sharing workflows [18], [19] (*anonymous* level). Table A1 in Section E maps each operational use case to the corresponding benchmark subsets and experimental sections in this paper.

TABLE A2
Internal collection and candidate recall of declared foreign keys (whole database / evaluable keys).

Family	DBs	Tables	Gold FKs	Cand. recall
BIRD	80	597	657	0.939 / 0.951
Spider	199	988	863	0.965 / 0.982
ReDeLEx	71	687	923	0.996 / 1.000
Samples	3	40	46	0.935 / 1.000
TPC	2	32	112	0.991 / 1.000

TABLE A3
Macro F1 on BIRD and Spider over three fold partitions: the pre-specified one (P0) and two further ones (P1, P2). Last row of each block: paired difference averaged per database over the partitions, with 95% CI.

Names	Method	P0	P1	P2	Mean
original	RefSel	0.897	0.899	0.894	0.897
original	GBDT	0.876	0.891	0.886	0.884
original	Fast-FK	0.827	0.841	0.837	0.835
original	Rostin-style	0.806	0.814	0.809	0.810
	RefSel − GBDT	+0.021	+0.008	+0.008	+0.012 [.002,.024]
anonymous	RefSel (full)	0.762	0.742	0.745	0.750
anonymous	GBDT (= rule-sel.)	0.715	0.728	0.733	0.725
anonymous	Fast-FK	0.649	0.641	0.655	0.648
	full − GBDT	+0.047	+0.014	+0.012	+0.024 [.007,.045]

H Extended Method Details

This section keeps the full-length method text that Section 3 of the main paper condenses.

H.1 Domain estimates and relation to MC-FK

Estimating the domain: Because the true generative domain $\mathcal{U}$ is not explicitly declared in a database, we compute four complementary domain-size estimates (detailed in Section B): (1) the global integer or date span covered jointly by $S_e \cup P_e$ (rng); (2) the local span strictly between the minimum and maximum values of S_e (loc); (3) the total

	names	customer_id → city_id	payment.staff_id →	declared found (of 9)	false edges (total)
RefSel	original	rejected	staff	9	0
	anonymous	rejected	staff	6	3
GBDT	original	rejected	staff	9	1
	anonymous	rejected	staff	6	3
Fast-FK	original	rejected	staff	8	2
	anonymous	rejected	store	2	6
MC-FK	original	accepted	both	7	47
	anonymous	accepted	both	7	47

Final-run predictions, restricted to the six tables. Green: correct decision, red: wrong, grey: none. The declared target of payment.staff_id is staff.

Fig. A1. Inclusion traps in the MySQL Sakila database. Nine declared foreign keys among six tables coexist with 72 undeclared exact unary inclusions into unique columns (81 exact inclusions in total). Two traps are highlighted: (1) *coincidental inclusion*, where all 599 `customer_id` values lie inside the 600 auto-increment `city_id` values; and (2) *value-equivalent targets*, where `payment.staff_id` $= \{1, 2\}$ matches both `staff.staff_id` (declared target) and `store.store_id`. The table compares final-run decisions of RefSel and three baselines (green: correct, red: wrong).

TABLE A4
Components on BIRD/Spider: paired test-fold difference in macro F1 with 95% CI on the pre-specified partition (Table A3 for the others), and whether the inner-validation rule admitted it. †: measured with the earlier candidate set. ‡: post hoc (Section 5.2).

Component	Original	Anonymous	Adopted
Signif. + stacking	+0.021 [.008,.035]	+0.047‡ [.025,.072]	orig. only
Signif. + stacking†	+0.018 [.005,.031]	+0.027 [.005,.048]	–
Stacking alone†	+0.018 [.006,.032]	+0.023 [.004,.043]	–
Significance alone†	−0.002 [−.007,.003]	+0.006 [−.008,.020]	no
Structural potentials†	−0.003 [−.010,.004]	+0.008 [−.006,.024]	no
Per-DB bias†	−0.001 [−.009,.006]	+0.003 [−.010,.016]	no
Hub decoding†	−0.039 [−.057,−.024]	+0.003 [−.008,.015]	no
Key-context features†	+0.001 [.000,.004]	+0.005 [.001,.010]	no

number of distinct values observed across all columns of the same atomic type in database D (typ); and (4) for string columns, the combinatorial size of the inferred character-class pattern (pat). Each estimate is lower-bounded by $|S_e \cup P_e| = N + k - m$, and Stage 2 summarizes them by their most conservative value, s_e^{cons} of Eq. (4), evaluated over the estimators applicable to the column type, alongside the individual estimates and the analytical KL bound of Lemma 1(b). As shown in Stage 2 of Fig. 1, when both source and target are dense auto-increment counters starting at 1, the local estimator yields $\hat{u} \approx N$ and therefore $s_e^{\text{cons}} \approx 0$, immediately exposing that the inclusion carries almost zero statistical evidence. Because real non-references can still exhibit non-uniform clustering not captured by H_0, s_e^{cons} and its components are fed into the Stage 2 classifier as continuous features rather than applied as a hard cutoff.

Relation to MC-FK: It is instructive to contrast Eq. (3) with MC-FK [11], which tests the opposite null hypothesis—namely, that the source values S_e behave like a uniform random sample drawn *from the target key* P_e—and accepts candidates that pass a goodness-of-fit test. Because any prefix range $\{1, \ldots, 599\}$ looks compatible with a slightly longer range $\{1, \ldots, 600\}$, MC-FK readily accepts the false link `customer.customer_id` → `city.city_id` in Sakila (Fig. A1), whereas our local domain estimator yields $s_e^{\text{cons}} \approx 0$. However, when a single source column is included in multiple targets that share the exact same value set (Fig. A1), no inclusion test alone can tell the targets apart; resolving or bounding that target competition requires our second theoretical principle.

H.2 Identifiability in practice

In practice, a learned classifier observes a feature vector rather than the raw instance O; Proposition A1 in Section A proves that the exact same $1/\nu$ upper bound holds whenever ν competing targets share identical feature vectors under a relabeling-invariant feature map. Crucially, however, stacked relational features that summarize how *other* tables in the schema interact with each candidate target can break this local symmetry when the surrounding schema is asymmetric.

Because real database tables are rarely perfectly isomorphic beyond their key columns, we use Proposition 1 and Corollary 1 in two complementary ways: as an explicit feature for symmetry breaking in Stage 2, and as the structural foundation for confidence calibration in Stage 4. Specifically, for every candidate $e \in g(e)$ we hash its target value set P_e to count its *value-equivalence class size* $\nu_e = |\{f \in g(e) : P_f = P_e\}|$. Having $\nu_e > 1$ is a necessary condition for target indistinguishability in the anonymous regime, though not sufficient: two target tables with identical key sets $\{1, 2\}$ may still differ in their non-key attributes or in how many other source groups reference them. Stage 2 therefore takes ν_e alongside stacked schema-context features to break value equivalence whenever surrounding schema evidence exists (evaluated in Section 5.4), while Stage 4 uses Corollary 1 to divide confidence among targets that remain near-tied after scoring (Section 3.5).

H.3 Stage 2 features

As depicted in Stage 2 of Fig. 1, the context scorer estimates the probability that each candidate $e \in \mathcal{E}(D)$ is a declared

TABLE A5
External benchmarks: candidate recall and macro F1 (whole database / evaluable keys). LLM-FK: our same-model rerun on the shared candidates (whole database) / the published value, from its own candidate set and data versions.

	Candidate recall		Original names		Anonymous	
Database	Whole	Evaluable	RefSel	LLM-FK (rerun / pub.)	RefSel	LLM-FK (rerun / pub.)
TPC-H	1.000	1.000	0.952 / 0.952	0.947 / 1.00	0.909 / 0.909	0.818 / 0.67
Northwind	0.846	1.000	0.870 / 0.952	0.917 / 1.00	0.800 / 0.870	0.750 / 0.92
TPC-E	0.918	1.000	0.683 / 0.718	0.887 / 0.93	0.745 / 0.778	0.646 / 0.71
AdventureWorks OLTP	1.000	1.000	0.849 / 0.849	0.918 / 0.94	0.671 / 0.671	0.492 / 0.53
MusicBrainz	0.581	1.000	0.594 / 0.806	0.714 / 0.95	0.428 / 0.575	0.162 / 0.76
AdventureWorks DW	1.000	1.000	0.755 / 0.755	–	0.539 / 0.539	–
TPC-DS	0.990	1.000	0.910 / 0.914	–	0.826 / 0.830	–

TABLE A6
With original names: whole-database F1 and number of LLM calls of RefSel with the bounded LLM prior and of the LLM-FK rerun with the same model and candidates.

	RefSel	+ LLM prior		LLM-FK rerun	
Database	F1	F1	calls	F1	calls
TPC-H	0.952	0.952	9	0.947	55
Northwind	0.870	0.917	6	0.917	56
TPC-E	0.683	0.869	32	0.887	211
AdventureWorks OLTP	0.849	0.926	155	0.918	3,495
MusicBrainz	0.594	0.696	644	0.714	9,688
Mean / total	0.790	0.872	846	0.877	13,505

TABLE A7
Choice of model (DeepSeek V4.1 Flash and V4 Pro): macro F1 with original names on LLM-FK's five benchmarks (sensitivity analysis) and on the nine conventional databases of the second independent round that may be sent to a model (post hoc), and LLM calls with V4 Pro on the benchmarks. "+ LLM" = bounded prior on all source groups and verifier veto.

	Benchmarks (5)		Independent (9)		Calls
Line	Flash	Pro	Flash	Pro	Pro, bench.
GBDT, no LLM	0.762		0.811		0
RefSel, no LLM	0.790		0.803		0
GBDT + LLM	0.890	0.875	0.899	0.893	1,045
RefSel + LLM	**0.892**	**0.880**	0.899	0.893	
LLM-FK (same model)	0.877	0.828	**0.906**	**0.901**	13,484
LLM-FK, anonymous	0.574	0.477	–	–	13,475

TABLE A8
Registered databases, evaluated once: whole-database / evaluable-key F1 of RefSel (rule-selected). DNF: not finished within the registered budget.

Database	Tables	Original	Anonymous
Oracle order entry	7	0.947 / 0.947	0.889 / 0.889
Flight booking	8	0.889 / 0.889	0.889 / 0.889
USDA nutrients	10	0.857 / 0.900	0.727 / 0.762
Oracle sales history	9	0.625 / 0.625	0.533 / 0.533
Dellstore	8	0.500 / 0.500	0.500 / 0.500
SAP SALT (unique targets)	4	1.000 / 1.000	1.000 / 1.000
SAP SALT (all relations)	4	0.500 / 0.500	0.500 / 0.500
OFBiz	857	0.464 / 0.786	0.202 / 0.393
Dolibarr	440	0.059 / 0.066	0.029 / 0.031
iDempiere	912	DNF (time)	0.064 / 0.113
MusicBrainz (full)	380	DNF (memory)	DNF (memory)

TABLE A9
CTU extension set (14 databases, registered before any run, evaluated once): macro F1, whole-database protocol.

Method	Original	Anonymous
Inclusion rule	0.649	0.601
MC-FK [11]	0.689	0.689
Fast-FK [12]	0.723	0.654
Rostin-style [13]	0.624	0.614
HoPF-style [14]	0.581	0.616
Auto-BI-lite [15]	0.571	0.575
GBDT context classifier	**0.810**	**0.700**
RefSel (rule-selected)	0.765	**0.700**

foreign key by combining three progressively broader feature families (enumerated in Section B): *edge features* $\boldsymbol{\xi}_e$, *group competition features* $\boldsymbol{\gamma}_e$, and *stacked relational features* $\boldsymbol{\alpha}_e$. First, *edge features* $\boldsymbol{\xi}_e$ characterize candidate e in isolation: its coverage θ_e, conservative inclusion significance s_e^{cons} and individual domain estimates (16 significance features in total), distinct-cardinality ratio k_e/N_e, distributional moments of source and target values, and—Strictly in the original naming condition—lexical string similarity between (A, X) and (B, Y). Second, *group features* $\boldsymbol{\gamma}_e$ capture how candidate e compares against competing targets within the same source group $g(e)$, including the relative rank of e by coverage θ_e and by target size N_e, the log-significance gap $\log(1 + \max_{f \in g(e)} s_f) - \log(1 + s_e)$ separating e from the group leader, and the value-equivalence class size ν_e.

Third, to capture schema-wide structural cues without expensive iterative message passing, *stacked relational features* $\boldsymbol{\alpha}_e$ summarize the first-pass predictions across the rest of the database. Specifically, a first-pass classifier f_1 maps local and group features to initial logits $z_e^{(1)} = f_1(\boldsymbol{\xi}_e, \boldsymbol{\gamma}_e)$ and edge probabilities $p_e = \sigma(z_e^{(1)})$, where $\sigma(z) = (1+e^{-z})^{-1}$. Writing $M_h = \max_{f \in h} p_f$ for the maximum first-pass probability in source group h, Stage 2 computes 20 stacked relational features for each candidate e, four primary examples of which are:

$$\text{share}_e = \exp(z_e^{(1)}) / \textstyle\sum_{f \in g(e)} \exp(z_f^{(1)}), \tag{A3}$$

$$\text{in}_e = \textstyle\sum_{f \notin g(e):\, (B_f, Y_f) = (B_e, Y_e)} p_f, \tag{A4}$$

$$\text{out}_e = \textstyle\sum_{h \neq g(e):\, A_h = A_e} M_h, \tag{A5}$$

$$\hat{\rho}_D = |\mathcal{T}_D|^{-1} \textstyle\sum_{h \in \mathcal{G}_D} M_h. \tag{A6}$$

Each quantity encodes a distinct structural intuition: share_e (Eq. (A3)) measures the softmax probability share of candi-

date e within its competing source group $g(e)$; in_e (Eq. (A4)) sums the first-pass probabilities of candidates from *other* source groups that point to the exact same target key (B_e, Y_e), capturing hub authority (true primary keys are often referenced by multiple tables, which helps break ties when $\nu_e > 1$); out_e (Eq. (A5)) estimates how many other foreign keys originate from source table A_e; and $\hat{\rho}_D$ (Eq. (A6)) estimates the overall foreign-key density per table in database D. To prevent self-reinforcement, all relational sums over external groups (A4)–(A5) strictly exclude the current group $g(e)$. A second-pass classifier f_2 then takes all three feature families to output the final edge logit $z_e = f_2(\boldsymbol{\xi}_e, \boldsymbol{\gamma}_e, \boldsymbol{\alpha}_e)$. This two-pass architecture instantiates stacked generalization [20], [21] for collective relational classification [22]; during training, the first-pass logits $z_e^{(1)}$ used to construct $\boldsymbol{\alpha}_e$ are computed strictly out-of-fold via group-disjoint cross-fitting so that f_2 never sees leaked training labels.

Both f_1 and f_2 are histogram gradient-boosted decision tree classifiers trained by minimizing the group-balanced binary logistic loss

$$\mathcal{L} = \sum_{D} \sum_{e \in \mathcal{E}(D)} \omega_e \, \ell\big(z_e, \mathbb{1}[e \in G_D]\big), \quad \omega_e \propto \frac{1}{|\mathcal{G}_D| \, |g(e)|}, \tag{A7}$$

where the inverse weighting by $|\mathcal{G}_D| \, |g(e)|$ ensures that every training database and every source group contributes equally to the gradient, preventing giant schemas or high-fan-out candidate groups from dominating the learned trees. Model hyperparameters and cross-validation protocols are detailed in Section 5.2.

H.4 Scope of the risk guarantee

Three practical aspects of Proposition 2 merit emphasis. First, the bound controls the expected false-discovery proportion averaged across new risk units drawn from the calibration distribution rather than guaranteeing $\mathrm{FDP}_{\hat{\tau}}(D) \leq \alpha$ on every single database in isolation, and its validity relies on defining risk units at the level of independent database families [5]. Second, strict finite-sample validity requires a *frozen program*: all learned components—Stage 1 rules, Stage 2 models (f_1, f_2), Stage 3 acceptance bias b and key scorer, Stage 4 calibrator ψ, and the threshold grid—must be locked before reading calibration labels. Because grouped cross-validation on the internal benchmark only approximates this separation (Section C), Section 5.6 executes a strictly frozen protocol: locking the entire model on development databases (**D**), calibrating $\hat{\tau}$ on a separate repository (**C**), and evaluating once on held-out independent test repositories (**T**). Third, while Proposition 2 guarantees risk control for *any* fixed ranking score, the choice of confidence score determines how much true output survives at tolerance α. In principle, because naive edge probability $\sigma(t_e)$ assigns high scores to ambiguous multi-target groups, LTT should have to raise $\hat{\tau}$ conservatively across all edges, whereas c_e pushes ambiguous edges down and should allow $\hat{\tau}$ to retain more unambiguous true references; empirically, however, Sections 5.5 and 5.6 do not establish a uniform retained-recall advantage for c_e over $\sigma(t_e)$ at fixed α.

H.5 LLM prior and verifier

Because multiple source groups are batched into a single prompt, the number of LLM API calls scales with the number of ambiguous source groups $|\mathcal{Q}(D)|$ rather than with the quadratic number of candidate pairs $|\mathcal{E}(D)|$. On our development databases, the selective margin $\mu = 5$ queries roughly half of all source groups; setting $\mu = \infty$ queries every group, which raised out-of-fold macro F1 on the 284 development databases from 0.897 to 0.911 and serves as the default setting for the frozen independent evaluation (Section 5.6). Every prompt contains strictly the schema metadata and sample rows of the database under evaluation, and responses are cached deterministically by SHA-256 prompt hash. We invoke the LLM components only under the original naming condition, as opaque identifiers in the anonymous condition strip away the lexical semantics that language models rely on.

Post-decoding verifier: After Stage 3 produces the decoded set $\hat{F}(D)$, the verifier reviews each selected edge $e \in \hat{F}(D)$ by presenting the LLM with both tables' schemas, five to ten sample rows of the participating columns, and the Stage 1 inclusion statistics (θ_e, k_e, N_e). Operating in batches of several edges per prompt, the LLM replies `yes`, `unsure`, or `no` with a brief rationale. Any decoded edge receiving a `no` verdict is removed immediately (*veto*), and for Stage 5 retention the confidence c_e of each surviving or evaluated edge is multiplied by 1, $\frac{1}{2}$, or 0.05 for `yes`, `unsure`, or `no`, respectively. Across the 284 development databases, the verifier vetoed 109 decoded edges, 100 of which were genuine false positives (91.7% precision on removals), and increased the area under the ROC curve (AUC) of c_e for separating true from false decoded edges from 0.77 to 0.79. Crucially, verifier API calls scale linearly with the compact set of decoded edges $|\hat{F}(D)|$ rather than with $|\mathcal{E}(D)|$.

I Extended Experimental Report

This section reproduces the full-length experimental report that Section 5 of the main paper condenses. Tables and figures that remain in the main paper are cited with their main-paper numbers; metrics are defined in Section 5.1 of the main paper.

I.1 Setup

Internal collection: Our internal development benchmark comprises 355 non-empty relational databases drawn from five distinct families: 80 from BIRD [23], 199 from Spider [24], 71 from ReDeLEx [25], [26], three canonical sample databases (e.g., Sakila), and two TPC benchmarks (shown in the left panel of Fig. 2 and detailed in Table A2 in Section F; seven completely empty Spider schemas are excluded). To prevent data leakage across copies or variants of the same underlying schema, we group related schemas into duplicate-free source groups and assign each intact group to a single fold across five outer cross-validation folds. All 355 databases are evaluated in the anonymous condition, and the 284 databases outside ReDeLEx are evaluated in the original naming condition. Because ReDeLEx stores pre-reindexed dense surrogate keys [26] that behave differently from naturally occurring identifiers, we report ReDeLEx separately in Table 1 and base our primary internal conclusions on the 279 BIRD and Spider databases.

External and pre-registered databases: As summarized in the right panels of Fig. 2, we evaluate out-of-distribution generalization on two tiers of external data. First, for direct comparison with prior literature, we test on the five benchmarks evaluated by LLM-FK [8] (TPC-H, Northwind, TPC-E, AdventureWorks OLTP, and MusicBrainz) plus two classic analytical schemas (AdventureWorks DW and TPC-DS), after purging all duplicate groups and a partial relative of AdventureWorks from the internal training pool (our dataset versions for MusicBrainz and TPC-E differ slightly from LLM-FK's; see Section F). Because these seven benchmarks were inspected during early model development—and informed one negative design decision, namely rejecting an unsupervised per-database bias shift—they do not constitute a strictly untouched test set. Second, for a truly blind confirmatory evaluation, we pre-registered ten additional databases (locking cryptographic SHA-256 hashes of the data, ground-truth labels, and committed model configuration prior to any execution) and evaluated them exactly once: five public schemas with declared constraints (Postgres Pro flight-booking, USDA nutrient data, Dellstore, Oracle sales-history, and Oracle order-entry), three large open-source ERP systems (Dolibarr, Apache OFBiz, and iDempiere), SAP SALT enterprise sales data, and the complete unsubsetted public MusicBrainz dump (380 tables and 774 declared foreign keys, of which 394 are evaluable because 160 tables are unpopulated in the public release). Finally, TPC-H and TPC-DS at scale factor 10 are used exclusively to measure runtime scalability.

Baselines: To isolate the quality of candidate scoring and global decoding from candidate generation, all methods operate on the exact same Stage 1 candidate set $\mathcal{E}(D)$ unless explicitly noted. We compare RefSel against seven baselines spanning four paradigms: (1) a naive *Inclusion rule* that selects the highest-coverage target in each source group; (2) the two closest name-free statistical methods, *MC-FK* [11] (with its earth-mover distance threshold tuned on training folds rather than on test labels) and *Fast-FK* [12] (with its unpublished heuristic feature weights tuned via grid search on training folds); (3) a supervised *Rostin-style* classifier [13] and a faithful reimplementation of *HoPF*'s holistic heuristics on shared candidates [14] (since no author code is publicly available); (4) *Auto-BI-lite*, which feeds our local candidate scores into the acyclic k-MCA-CC graph objective of Auto-BI [15] to isolate the effect of acyclic vs. cyclic global decoding; and (5) a strong *GBDT context classifier* trained on the exact same candidate set and Stage 3 decoder, but ablated of inclusion significance features (s_e), two-pass relational stacking ($\boldsymbol{\alpha}_e$), and identifiability-aware confidence (c_e). All baseline hyperparameters are tuned strictly on inner validation folds, with the baseline GBDT classifier receiving an even larger hyperparameter search grid than RefSel (Section F). On external and independent benchmarks, we additionally execute *LLM-FK* [8] using its published multi-agent prompts and the exact same backbone LLM as RefSel, reporting both our same-model rerun and LLM-FK's originally published numbers.

Evaluation protocol and pre-specified configurations: We report macro F1, pooled precision, pooled recall, and Stage 1 candidate recall under the denominators defined in Section 5.1. Statistical significance of paired differences is assessed via two-sided Wilcoxon signed-rank tests over duplicate-group means with Holm step-down multiple-testing correction within each comparison block (defined by a naming condition and benchmark collection), accompanied by 95% bootstrap percentile confidence intervals over 2,000 group-level resamples. While Table 1 reports results on the primary fold partition fixed prior to final execution, we also repeated the entire internal cross-validation across two additional random group-disjoint fold partitions to quantify partition variance (Table A3 in Section F).

To guard against overfitting during ablation design, we committed to a strict component-admission rule per naming condition before running test folds: an architectural component (significance + relational stacking) enters the *rule-selected* configuration of RefSel if and only if its paired F1 gain on inner validation folds has a 95% confidence interval that excludes zero. Under original names, inner validation passed this threshold, so *RefSel (rule-selected)* and *RefSel (full)* (which activates all components) are identical. Under the anonymous condition, the inner-validation confidence interval slightly straddled zero, causing *RefSel (rule-selected)* to fall back to the base GBDT scorer without significance or stacking. Because *RefSel (full)* subsequently achieved substantially higher accuracy on the internal anonymous test folds (0.762 vs. 0.715), we locked *RefSel (full)* as a second tracked configuration prior to running any external, pre-registered, or independent round-$1/2/3$ benchmarks. Consequently, anonymous internal test-fold numbers for RefSel (full) are marked with ‡ as post hoc in Table 1, whereas all of its external, registered, and independent evaluations are strictly confirmatory. Unless explicitly labeled "(full)", anonymous internal numbers refer to the conservative rule-selected configuration.

Implementation: RefSel is implemented in Python 3.12 using DuckDB 1.5 for SQL semi-join profiling, scikit-learn 1.9 for histogram gradient-boosted trees, and OR-Tools CP-SAT 9.15 [2] for integer linear decoding. All language-model experiments use DeepSeek V4.1 Flash at temperature 0 with deterministic SHA-256 prompt caching, executed on an Apple M2 workstation (8 cores, 24 GB RAM).

I.2 RQ1: Effectiveness

Table 1 summarizes five-fold cross-validation performance on the internal benchmark, while Table A3 in Section F details stability across fold partitions. In the *original naming condition* on BIRD and Spider, RefSel achieves a macro F1 of 0.897 (pooled precision 0.838, pooled recall 0.863), outperforming every statistical, heuristic, and learned baseline. On the pre-specified partition (P0), RefSel exceeds the ablated GBDT context classifier on the exact same candidates by $+0.021$ F1 (95% CI $[0.008, 0.035]$, Holm-corrected $p = 0.004$), driven primarily by suppressing false-positive inclusions at matched recall (analyzed in Section I.3). Because partition P0 is somewhat favorable to RefSel, repeating the evaluation across two additional random partitions (P1 and P2, Table A3) narrows the gap: averaged across all three partitions, the paired gain over GBDT is $+0.012$ (95% CI $[0.002, 0.024]$), with individual duplicate groups favoring each method nearly equally often (47 wins vs. 48 losses; Wilcoxon $p = 0.25$). Thus, when descriptive column names are present, the incremental gain

of significance and stacking over a well-tuned GBDT + Stage 3 decoder is modest and concentrated in a subset of schemas—primarily in Spider ($+0.015$, CI $[0.003, 0.030]$) rather than BIRD ($+0.005$). By contrast, RefSel's lead over published methods is large and partition-invariant, beating Fast-FK by $+0.062$ (CI $[0.036, 0.088]$), Rostin-style by $+0.091$, and Auto-BI-lite by $+0.128$.

In the *anonymous condition*, baselines that lean heavily on lexical similarity suffer severe drops—Rostin-style plunges from 0.806 to 0.543 and Fast-FK drops from 0.827 to 0.649—whereas purely distributional MC-FK remains flat at 0.617. Here, RefSel (rule-selected) coincides with the GBDT context classifier at 0.715 macro F1, outperforming Fast-FK by $+0.066$ (CI $[0.033, 0.098]$) and Auto-BI-lite by $+0.082$. Activating inclusion significance and relational stacking in RefSel (full) raises anonymous macro F1 on partition P0 to 0.762 ($+0.047$ over GBDT, post hoc); averaged across all three partitions (Table A3), RefSel (full) maintains a statistically significant gain of $+0.024$ over GBDT (95% CI $[0.007, 0.045]$, Wilcoxon $p = 0.007$), though the gain on partitions P1 ($+0.014$) and P2 ($+0.012$) individually falls below single-partition significance. Finally, on ReDeLEx (rightmost columns of Table 1), where pre-reindexed surrogate keys reduce range collisions, both RefSel configurations achieve ≈ 0.745 macro F1 versus 0.571 for the best baseline.

Progressive name degradation and distractor tables: Fig. 3 examines robustness across intermediate lexical degradations (left panel) and structural noise (right panel) on 274 BIRD and Spider schemas. Tracing the left panel of Fig. 3 from left to right, stripping vendor prefixes (*prefix* level) reduces RefSel's macro F1 negligibly ($0.897 \rightarrow 0.893$), applying heavy SAP-style vowel-and-token compression (*abbreviation* level) costs roughly three points (0.865), and the bulk of the drop to the *anonymous* level (0.715) occurs only when lexical cues vanish completely. Across the baselines, name-free MC-FK and the inclusion rule remain flat, Fast-FK loses less than one point under abbreviation ($0.827 \rightarrow 0.820$), GBDT and Auto-BI-lite mirror RefSel's three-point abbreviation dip, and heavily lexical Rostin-style ($0.806 \rightarrow 0.751$) and HoPF-style ($0.759 \rightarrow 0.657$) degrade steepest early on.

The right panel of Fig. 3 injects one to three synthetic *distractor tables* into every database—each populated with an auto-increment key, two to three integer columns sampled uniformly inside existing key ranges, a low-cardinality status flag, and two text attributes. Although zero true foreign keys touch these distractor tables, their integer columns nest inside existing primary keys by construction, creating a burst of coincidental inclusions. Consequently, RefSel's macro F1 drops from 0.897 to 0.782 (original) and from 0.715 to 0.544 (anonymous), with 56% of all false positives involving a distractor table. Baselines that rely strongly on value containment suffer equally or worse: with original names, Fast-FK, HoPF-style, and MC-FK plunge by 15–16 points and the naive inclusion rule drops by 26 points, whereas classifiers that weight lexical features heavily over value inclusion drop less (GBDT $0.876 \rightarrow 0.790$, Rostin-style $0.806 \rightarrow 0.795$, Auto-BI-lite $0.769 \rightarrow 0.741$, leaving all three level with or slightly above RefSel under named distractors); in the anonymous condition, every method loses 15–25 points and none exceeds RefSel's 0.544. Crucially, as shown by the callout boxes in Fig. 3 (right), injecting uncalibrated distractors also acts as a severe covariate shift for Stage 5 risk control: at target risk $\alpha = 0.2$, the realized database-mean FDP rises from 0.12 to 0.27 (original) and from 0.13 to 0.38 (anonymous), whereas at the stricter $\alpha = 0.1$ level it remains safely controlled at 0.04 in both regimes. This previews the exact failure mechanism observed on sparse ERP demonstration schemas in Section I.8.

Cross-family transfer and end-to-end profiling: When an entire benchmark family is held out from training, RefSel trained strictly on Spider transfers to BIRD at 0.788 (original) and 0.729 (anonymous) macro F1, while RefSel trained strictly on BIRD transfers to Spider at 0.841 (original) and 0.633 (anonymous); anonymous transfer to Spider is lower because Spider's compact schemas harbor the highest concentration of value-equivalent target keys (Section I.3). Finally, because our shared candidates are generated by Stage 1 of RefSel, RefSel's end-to-end accuracy equals Table 1; by contrast, running Fast-FK end-to-end with its own native candidate generator (tuned over a reduced weight grid) yields only 0.556 (original) and 0.557 (anonymous) across all internal databases.

I.3 RQ2: Mechanism

Error taxonomy and where recall is lost: To understand what drives the F1 numbers in Table 1, Table 2 decomposes all predictions on BIRD and Spider into true positives (*TP*), *wrong targets* (where the source column list is indeed a declared foreign key, but the method selects the wrong competing target in $g(e)$), *missed references* (false negatives, further partitioned into *no Stage 1 candidate / pruned by key-consistency constraint (C2) / scored below threshold*), and *spurious references* (false positives on source columns that are not foreign keys at all). Stage 1 candidate generation misses only 70 of the 1,520 declared foreign keys (4.6%), confirming high candidate recall. Likewise, the Stage 3 decoding constraints (C1)–(C4) sacrifice almost zero achievable recall: under an oracle scorer, enforcing (C1)–(C4) retains all but 17 of the 2,523 internal gold foreign keys present in $\mathcal{E}(D)$ (10 of those 17 reference columns that contain duplicate values in the instance) and retains 100% of the 744 evaluable foreign keys across the seven external benchmarks. Comparing rows in Table 2 reveals how the methods differ: under original names, RefSel surpasses GBDT primarily by cutting spurious false alarms (from 222 down to 196) and reducing valid foreign keys lost to wrong primary-key selection under (C2) (from 33 down to 20). When moving to the anonymous condition, the dominant error shift is an explosion in *wrong targets* (jumping from 57 with names to 230 without names), and 177 of those 230 wrong-target errors (77%) occur between competing target tables whose key value sets P_e are completely identical. Activating significance and relational stacking in RefSel (full) recovers 64 missed references (reducing missed from 264 to 200) and trims both wrong targets and spurious edges.

Empirical verification of the identifiability bound: The right panel of Fig. 4 directly tests Proposition 1 by stratifying decoded source groups according to the number ν of competing targets in $g(e)$ that share the exact same value set ($\nu = 1, 2, 3, \geq 4$). In the anonymous condition (solid curves in Fig. 4, right), the fraction of decoded source

groups in which the selected target matches the declared gold target is 0.95, 0.59, 0.46, and 0.23 for $\nu = 1, 2, 3, \geq 4$, closely tracking the theoretical random-guessing ceiling $1/\nu = 1.00, 0.50, 0.33$, and 0.18 (dashed curve); for every $\nu \geq 2$, the 95% confidence interval contains $1/\nu$. Thus, without column names, value-equivalent targets are empirically resolved near the chance limit dictated by Proposition 1. In contrast, when original names are available (top curve in Fig. 4, right), lexical alignment breaks the symmetry and target-selection accuracy stays high at 0.98, 0.95, 0.66, and 0.99 across the same ν strata. Because 17% of all declared foreign keys in BIRD and 31% in Spider face at least one value-equivalent rival ($\nu \geq 2$), Proposition 1 places a hard ceiling on what *any* name-free method can achieve and explains why anonymous accuracy is systematically lower on Spider than on BIRD. While whole tables are less symmetric than their key columns alone—a classifier trained on label-free table-level summary statistics (column counts, string-column ratios, high-cardinality text indicators, and UCC counts) identifies the true target table in 0.68 of value-equivalent classes versus 0.43 by chance—forcing the Stage 3 decoder to route all references within an equivalence class to a single "hub" table fails to improve F1 (Table A4 in Section F): even an oracle that reveals the most-referenced table per equivalence class gains only +1.8 F1 points, because in over one-third of multi-target classes, different foreign keys genuinely reference different tables within the same equivalence class. This finding justifies our design choice to reflect unresolvable target competition in the Stage 4 confidence c_e rather than imposing hard hub constraints in Stage 3.

Diagnostic power of inclusion significance: Raw inclusion coverage θ_e barely separates true foreign keys from false candidates (achieving an AUC of just 0.52 on BIRD). By contrast, the left panel of Fig. 4 demonstrates that conservative inclusion significance s_e^{cons} (Eq. (4)) behaves exactly as Lemma 1 predicts across different key distributions: (1) on *sparse integer keys* ($u \gg N$), s_e^{cons} cleanly separates true references from coincidental overlaps with an AUC of 0.83; (2) on *dense auto-increment integer keys starting at* 1 ($u \approx N$), it assigns $s_e^{\text{cons}} < 1$ to 87% of true references and 98% of false candidates, correctly signaling that dense prefix inclusion carries zero statistical evidence either way; and (3) on *string keys*, s_e^{cons} provides no separation (AUC 0.46), because false string inclusions are rarely random character collisions under H_0, but rather distinct columns populated from the same shared vocabulary (such as ISO country or status codes)—which represents value-equivalent target ambiguity rather than random coincidence. On ReDeLEx, whose synthetic surrogate keys invalidate global type and pattern domain estimators, using significance in isolation increases false positives. Consequently, s_e^{cons} functions best as an analytical diagnostic of when value containment is trustworthy: ablating significance alone changes end-to-end F1 negligibly once relational stacking is present (Table A4), because gradient-boosted trees already approximate local domain ratios from $(\theta_e, k_e/N_e)$ and range statistics, while group-level significance gaps feed the second-pass stacked features.

Component ablations and negative results: Table A4 in Section F reports paired test-fold F1 gains and inner-validation admission decisions for every architectural component we explored. Combining inclusion significance with two-pass relational stacking delivers our largest internal gain (+0.021 original, +0.047 anonymous on P0), with relational stacking accounting for the vast majority of the improvement. Equally important, we document four plausible extensions that yielded negative or negligible results and were excluded from RefSel: (1) *learned structural graph potentials* added to the Stage 3 decoder objective (trained via a structured perceptron to reward tightest nested targets per Lemma A1 and penalize transitive shortcuts; Section E) eliminated wrong targets but sacrificed an equal amount of recall, and the learned weight for tightest containment flipped sign in the anonymous regime where dense keys nest by default (−0.003 original, +0.008 anonymous); (2) an *unsupervised per-database prior-shift bias* estimated via the EM algorithm of Saerens et al. [27] produced no internal gain (−0.001) and degraded accuracy on external benchmarks; (3) *hub-consistent decoding* over value-equivalent targets hurt original-name F1 by −0.039 for the reason explained above; and (4) adding *inbound-inclusion context features* to the primary-key classifier improved key accuracy by less than one point (+0.001 to +0.005 F1) because our base key scorer already operates near oracle accuracy.

I.4 RQ3: Risk control

Cross-validated risk control: We first evaluate Stage 5 risk control under five-fold cross-validation on BIRD and Spider, where the group calibrator $\hat{\pi}_g$ and LTT threshold $\hat{\tau}$ are fitted on inner validation folds of each training split (an approximate protocol analyzed in Section C; Section I.5 next tests the exact frozen protocol on independent repositories). Table A10 and Fig. 5 summarize both the number of true foreign keys retained per database (top panel of Fig. 5) and the realized group-mean false-discovery proportion (bottom panel of Fig. 5) across target levels $\alpha \in \{0.05, 0.10, 0.20\}$. On the internal test folds, the realized group-mean FDP stays strictly below the requested ceiling α at every operating point: at $\alpha = 0.20$, realized FDP is 0.112 (original) and 0.126 (anonymous, rule-selected), and at $\alpha = 0.10$ it is 0.04 and 0.03 (across the two additional fold partitions P1 and P2, realized FDP at $\alpha = 0.20$ remains equally stable at 0.11–0.12 original and 0.13–0.14 anonymous). As emphasized in Section 3.5, Proposition 2 bounds the *expectation across risk units* rather than worst-case individual schemas: at $\alpha = 0.20$, 57 of the 279 individual databases exceed an FDP of 0.20 on their own while the population mean stays at 0.112.

When deployed outside the calibration distribution, however, the threshold $\hat{\tau}$ calibrated on internal folds holds only when the target domain matches the calibration distribution (Fig. 5, bottom): under whole-family holdout, risk control holds when transferring to Spider and to anonymous BIRD, but slightly breaches $\alpha = 0.20$ on named BIRD (0.22). On the seven external benchmarks, risk is well controlled (0.16 original and 0.07 anonymous at $\alpha = 0.20$). By contrast, on the ten pre-registered databases, realized FDP at $\alpha = 0.20$ reaches 0.36 in both naming conditions—driven by the sparse ERP demonstration schemas (0.59 original and 0.63 anonymous), whereas the five public schemas plus SAP SALT stay near target (0.28 and 0.22); at the stricter $\alpha = 0.10$ level,

TABLE A10
Correct foreign keys retained per database and realized group-mean FDP on the test folds (BIRD/Spider), by requested level α.

		Retained correct / DB			FDP
Names	Confidence	α=.05	.1	.2	at .2
original	GBDT, $\sigma(t_e)$	0.31	2.70	4.66	0.115
original	RefSel, $\sigma(t_e)$	0.54	2.43	4.70	0.112
original	RefSel, c_e	0.53	2.18	4.70	0.112
anonymous	RefSel, $\sigma(t_e)$	0.10	1.01	2.74	0.148
anonymous	RefSel, c_e	0.05	0.58	2.76	0.126
anonymous	RefSel (full), c_e	0.04	1.72	3.14	0.116

realized FDP on the registered set is 0.09 and 0.00, because the high threshold abstains on almost all edges.

Comparing identifiability-aware confidence c_e against raw edge probability $\sigma(t_e)$: Fig. 6 plots the anonymous precision–recall curves and marks the operating points selected at $\alpha \in \{0.05, 0.10, 0.20\}$ when thresholding c_e (Eq. (6)) versus raw sigmoid probability $\sigma(t_e)$. Because c_e and $\sigma(t_e)$ rank the exact same decoded edges $\hat{F}(D)$ differently, neither uniformly dominates across all regimes: across 20 paired comparisons of retained true foreign keys per database (spanning internal test folds, held-out families, external benchmarks, and registered databases under both naming conditions at $\alpha \in \{0.10, 0.20\}$), c_e retains more true edges in 7 settings, $\sigma(t_e)$ wins in 5, and the two tie in 8. At $\alpha = 0.20$ with original names, both scores retain 99.9% of all decoded edges (4.70 correct FKs/DB in Table A10). At the stricter tolerance $\alpha = 0.10$, however, c_e retains slightly fewer true foreign keys on internal test folds: -0.26 per database (95% CI $[-0.45, -0.10]$) with original names and -0.43 (CI $[-0.66, -0.26]$) without names. This behavior follows directly from the tie-splitting factor $\hat{q}(e \mid g)$ in Eq. (6): whenever $\nu \geq 2$ competing targets tie, $c_e \leq \hat{\pi}_g/\nu \leq 0.50$, which pushes all edges from ambiguous groups below strict thresholds ($\hat{\tau} > 0.50$) even when the decoder happened to pick the right target. Conversely, in the anonymous condition at $\alpha = 0.20$, c_e achieves a lower realized FDP than $\sigma(t_e)$ (0.126 vs. 0.148) while retaining slightly more correct foreign keys (2.76 vs. 2.74 per database). Under original names at $\alpha = 0.10$, GBDT with $\sigma(t_e)$ retains the most (2.70/DB). Ultimately, improving the underlying scorer yields a much larger retention boost at fixed risk than swapping the confidence formula: in the anonymous condition, RefSel (full) with c_e retains 3.14 true FKs/DB at $\alpha = 0.20$ (versus 2.76 for rule-selected) and 1.72 at $\alpha = 0.10$ (versus 0.58). Meanwhile, the isotonic group probability $\hat{\pi}_g$ achieves an expected calibration error (ECE) between 0.02 and 0.11 across benchmark families, though sparsely populated mid-range confidence bins can deviate by up to 0.30.

I.5 RQ5 (Round 1): Independent calibration and test

Strictly frozen three-way split protocol: To evaluate the frozen-program calibration-and-test design motivating Proposition 2—without cross-validation overlap—we partitioned databases by repository origin into three strictly disjoint sets that share zero schemas or duplicate groups: (1) the *development part* consists of the 283 internal databases (270 source groups) remaining after removing the entire ReDeLEx family; every learned parameter—Stage 1 rules, Stage 2 tree ensembles (f_1, f_2) and acceptance bias b, Stage 3 key models, Stage 4 isotonic map $\hat{\pi}_g$ (fit on development out-of-fold predictions), and the 20-point quantile threshold grid—was trained exclusively on this development pool, and cryptographic hashes of all model weights and inference code were sealed before reading any calibration or test label; (2) the *calibration part* comprises 29 databases (24 independent source groups) from the CTU relational repository [28] (made completely disjoint from development once ReDeLEx, a port of CTU, was removed, and filtered by strict same-origin fingerprint rules in Section B to exclude any overlap with external, registered, or Table A9 schemas); the frozen program executes once on the calibration part to select $\hat{\tau}$ via fixed-sequence Hoeffding–Bentkus testing at $\delta = 0.10$ (Section C); and (3) the *test part* comprises nine databases (nine independent source groups) drawn from two repositories never touched anywhere earlier in our study: six from RelBench (`avito`, `event`, `hm`, `trial`, `arxiv`, `ratebeer`) [29] and three from 4DBInfer (`diginetica`, `retailrocket`, `outbrain-small`) [30], whose primary keys are recoded as dense $0/1$-based row indices (the exact dense-integer regime analyzed in Section 3.2). Test evaluation was executed once using the sealed predictions and CTU-calibrated thresholds $\hat{\tau}$, and every metric was independently verified from the sealed files via a standalone NumPy script. We compare three lines sharing identical candidates, decoder, and LTT calibration: *GBDT with* $\sigma(t_e)$, *RefSel with* $\sigma(t_e)$, and *RefSel with* c_e (where RefSel denotes the full configuration fixed prior to calibration). We pre-registered a single primary endpoint: retained correct foreign keys per database at $\alpha = 0.20$ for *RefSel with* c_e versus *GBDT with* $\sigma(t_e)$ (tested via two-sided Wilcoxon signed-rank tests over test source-group means with Holm correction across the two naming conditions; $\alpha = 0.20$ was chosen because with $n = 24$ calibration units, the finite-sample Hoeffding–Bentkus bound at $\alpha = 0.10$ can reject only at near-zero empirical loss). A separate post-decoding classifier for ranking decoded edges achieved lower development AUC than $\sigma(t_e)$ (0.70 vs. 0.77) and was excluded prior to sealing.

Results of Round 1: Table A11 and Fig. A2(a) report the one-time Round 1 results. First, every reported test FDP point estimate was below the selected tolerance: on the test part, every line achieved a realized group-mean FDP of 0.00 at both $\alpha = 0.10$ and $\alpha = 0.20$ (with $n = 9$ test groups, the 90% Hoeffding–Bentkus upper confidence bound on true risk is 0.23). Second, however, our pre-registered primary comparison went decisively *against* RefSel with c_e: at $\alpha = 0.20$ under original names, RefSel with c_e retained 1.89 correct foreign keys per database compared with 6.33 for GBDT with $\sigma(t_e)$ (paired difference -4.4, 95% CI $[-7.4, -1.9]$, 0 wins vs. 6 losses, Wilcoxon $p = 0.03$), and under anonymous names it retained 0.00 versus 2.00.

Inspecting the calibrated thresholds $\hat{\tau}$ reveals why: on the 24 CTU calibration groups, GBDT's highest-scoring predictions happened to contain zero errors against declared constraints, allowing LTT to relax GBDT's threshold down to $\sigma(t_e) \geq 0.9996$; by contrast, with $n = 24$ calibration units at $\alpha = 0.20$ and $\delta = 0.10$, the Hoeffding–Bentkus test tolerates

TABLE A11
First independent round: correct foreign keys retained per database on the test part ($n = 9$ source groups) and realized group-mean FDP, by line and level; thresholds calibrated on 29 CTU databases (24 groups). Last column: decoded-set macro F1 on the calibration part (C) and the test part (T), computed after the run.

Names	Line	α=.1	.2	FDP at .2	F1 C / T
original	GBDT, $\sigma(t_e)$	3.78	6.33	0.00	0.675 / 0.952
original	RefSel, $\sigma(t_e)$	0.56	2.33	0.00	0.799 / 0.966
original	RefSel, c_e	0.00	1.89	0.00	0.799 / 0.966
anonymous	GBDT, $\sigma(t_e)$	0.56	2.00	0.00	0.698 / 0.878
anonymous	RefSel, $\sigma(t_e)$	0.67	0.67	0.00	0.725 / 0.905
anonymous	RefSel, c_e	0.00	0.00	–	0.725 / 0.905

an empirical calibration FDP of approximately 0.0487, so just two high-confidence false positives on small CTU schemas halted RefSel's threshold walk early. To understand those calibration errors, we audited all 10 high-confidence calibration edges that triggered threshold stopping against the underlying database instances (Section D): six of the ten "errors" are in fact genuine semantic foreign keys that the CTU schemas failed to declare in SQL metadata (such as `teams.franchID` → `teamsfranchises` in Lahman baseball and `train.(Date,Store)` → `features` in Walmart), one is a value-equivalent rival target, and only three are genuine errors (all occurring in the anonymous regime via coincidental inclusions: a reversed 1-to-1 key, a binary $0/1$ flag included in a 7-row lookup table, and overlapping surrogate counters). In a post hoc sensitivity analysis that treats the six undeclared CTU references as true positives, retained recall rises for all lines and the named gap shrinks to -1.3 FKs/DB (CI $[-3.4, +0.1]$, $p = 0.28$), whereas in the anonymous regime the three genuine errors still pin RefSel's threshold high (widening the anonymous gap from -2.0 to -5.0).

Crucially, the unthresholded Stage 3 decoded sets $\hat{F}(D)$ (last column of Table A11) tell the opposite story from the thresholded retained sets: on the sealed predictions, frozen RefSel achieved higher decoded macro F1 than frozen GBDT on both independent cohorts and under both naming regimes—scoring 0.799 vs. 0.675 (original; 10 wins, 3 losses, $p = 0.08$) and 0.725 vs. 0.698 (anonymous) on the 29 CTU calibration databases, and 0.966 vs. 0.952 (original) and 0.905 vs. 0.878 (anonymous) on the nine test databases. Because decoded F1 was examined after unsealing Round 1, we report it as post hoc here and elevated it to the pre-registered primary endpoint of Round 2 (Section I.6). Round 1 thus establishes two structural lessons: (1) the tie-penalized confidence c_e is overly conservative when Stage 2 context features have already broken target ambiguity (as occurs on recoded RelBench/4DBInfer keys), and (2) finite-sample LTT calibration with only $n = 24$ units is so strict (tolerating empirical FDP up to approximately 0.0487 at $\alpha = 0.20$ and 0.00185 at $\alpha = 0.10$; Section C) that missing ground-truth declarations in the calibration set dominate threshold selection.

I.6 RQ5 (Round 2): Independent evaluation of data-only vs. bounded-LLM pipelines

Pre-registered design of Round 2: Informed by Round 1, we pre-registered a second untouched test cohort comprising 41 databases across 10 independent source groups, drawn from repositories never used in development, calibration, or Round 1: four local databases from Spider 2.0 [31], 30 databases sampled from WikiDBs [32] (grouped as one source unit), the MIMIC-IV clinical demo [33], the Acquire Valued Shoppers benchmark from 4DBInfer, and five public tutorial schemas with declared constraints (KaggleDBQA Pesticide, BikeStores, pgexercises, PetClinic, and SQL Murder Mystery), after filtering out any schema sharing lineage with prior sets (such as Oracle HR, pubs, world, classicmodels, TPC-C, or Synthea). We reused the exact frozen models from Round 1 (re-verifying file hashes) and the same 29 CTU calibration databases.

Our pre-registered primary endpoint for Round 2 tested whether the unthresholded decoded macro F1 of *data-only RefSel* exceeds *data-only GBDT* (evaluated via one-sided Wilcoxon signed-rank tests over source-group means with Holm correction across naming conditions, treating retention as secondary). Crucially, under original names, we also pre-registered two secondary lines incorporating the bounded language-model mechanisms of Section 3.6 (developed strictly on the internal development pool): (1) the *bounded LLM prior* applied across all source groups ($\mu = \infty, B = 6$, which lifted development out-of-fold F1 from 0.897 to 0.911), and (2) the *post-decoding LLM verifier*, which inspects each decoded edge alongside sample rows and inclusion statistics, vetoes edges judged `no` (lifting development F1 from 0.897 to 0.902 by vetoing 100 false positives out of 109 removals), and scales retention confidence by $\{1, \frac{1}{2}, 0.05\}$ for {`yes`, `unsure`, `no`}. To keep retention lines directly comparable, the verifier was applied to GBDT's decoded edges as well, whereas the prior was pre-registered on RefSel (two alternative post-decoding confidence formulas that down-weighted $s_e \approx 0$ edges were rejected on development data prior to Round 2 because they penalized true dense-integer keys). Two restricted-license schemas (Acquire Valued Shoppers and MIMIC-IV) were excluded from external LLM API calls, and LLM-FK was executed with the exact same backbone LLM on shared candidates as an external baseline, with all predictions sealed before reading any test labels.

Results of Round 2: Table A12 and Figs. A2(b)–(c) present the one-time Round 2 evaluation, with every metric verified by independent recomputation (an initial automated run in which an API budget-ledger mismatch silently dropped all LLM calls was discarded prior to inspecting any outputs and documented in the sealed archive). First, our primary data-only hypothesis was *not* supported: without LLM evidence, frozen RefSel and frozen GBDT are statistically indistinguishable on the 41 test databases, achieving macro F1 of 0.438 vs. 0.456 under original names (difference -0.018, 95% CI $[-0.022, 0.008]$, one-sided $p = 0.78$) and 0.391 vs. 0.400 under anonymous names ($p = 0.25$), splitting wins and losses nearly evenly (10 vs. 11 original; 13 vs. 14 anonymous). Synthesizing Round 1 ($+0.12$ post hoc on CTU-29, $+0.01$ on RelBench/4DBInfer-9), Round 2 (-0.02), and the CTU-14 extension set (-0.05,

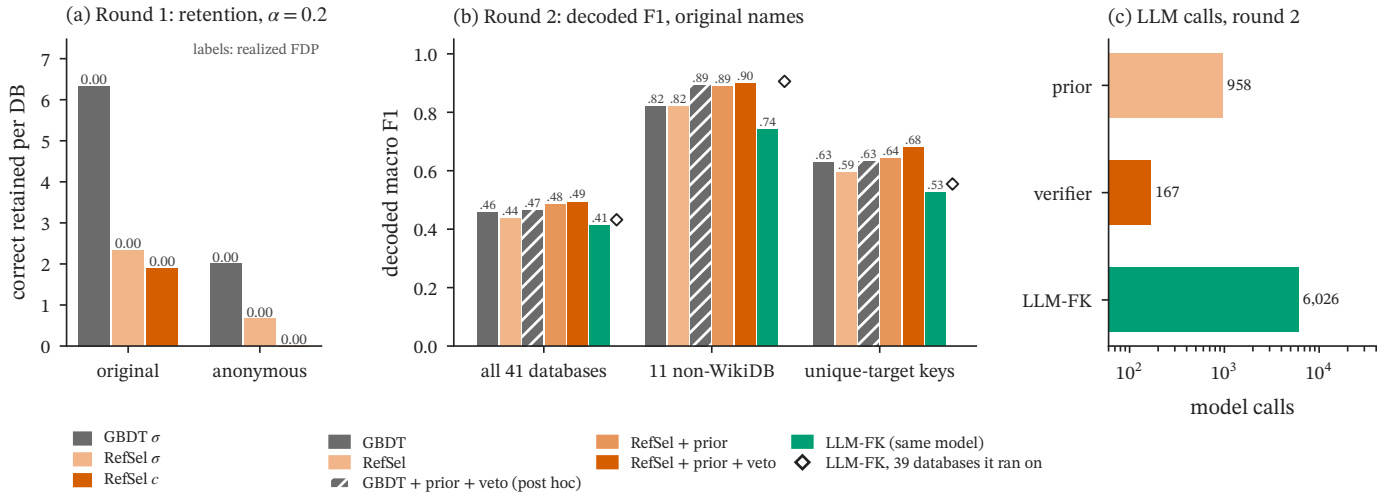


Fig. A2. Independent rounds. (a) First round, test part: correct foreign keys retained per database at $\alpha = 0.2$ for three retention lines with original and anonymous names; the number above each bar is the realized false-discovery proportion. (b) Second round: decoded-set macro F1 with original names over all 41 databases, the 11 non-WikiDB databases, and declared keys with unique targets. Hatched bars are post hoc (the prior applied to the GBDT classifier, which never received it in the frozen program), and the last two scopes are post hoc as well. LLM-FK counts its two excluded databases as empty; the diamond is its value on the 39 databases it ran on. (c) Model calls in the second round with original names; the verifier calls are shared by the RefSel and GBDT lines.

Section I.8), we conclude transparently that the data-only scorer refinements (significance + relational stacking) do not yield a confirmed F1 advantage over the base GBDT context scorer outside the BIRD/Spider development distribution.

Second, in sharp contrast, the *bounded language-model components* deliver a clear, statistically significant improvement (Fig. A2(b)): under original names, adding the bounded LLM prior across source groups lifts RefSel from 0.438 to 0.485 ($+0.047$, $p = 0.03$), and adding the post-decoding verifier veto raises it further to 0.493—outperforming the data-only GBDT baseline by $+0.037$ F1 (95% CI $[0.024, 0.124]$, $p = 0.016$, winning on 21 databases vs. 6 losses). Moreover, a post hoc cross-check (hatched bars in Fig. A2(b)) that re-decodes GBDT's logits using the exact same cached LLM prior and verifier responses (zero additional API calls) lifts GBDT to 0.459 (+prior), 0.467 (+veto), and 0.465 (+prior+veto); compared against GBDT+prior+veto, RefSel+prior+veto is $+0.028$ higher with a confidence interval crossing zero ($p = 0.50$, differing on only 3 of 10 source groups; on the development pool, the prior similarly lifts GBDT by $+0.031$ vs. $+0.014$ for RefSel, leaving the two scorers within $+0.004$, $p = 0.50$). This demonstrates that *bounded LLM evidence is scorer-agnostic*: it boosts either statistical scorer by roughly the same margin, showing that the architectural pipeline—Stage 1 candidates, Stage 3 collective decoding, the bounded LLM prior, the post-decoding verifier, and Stage 5 risk control—rather than the data-only scorer refinements, drives out-of-repository accuracy.

Comparing against *LLM-FK* on the same candidate pool and backbone LLM (Table A12 and Figs. A2(b)–(c)), on the 39 license-permitting databases where LLM-FK could run, LLM-FK reaches 0.432 (original, diamond in Fig. A2(b)) and 0.326 (anonymous) versus 0.473 and 0.368 for RefSel (or 0.411 and 0.310 across all 41 databases when counting the two license-excluded databases as empty outputs; the matched 39-database comparisons use 8 source groups and are not statistically significant; among the all-41 comparisons using 10 groups, only the anonymous comparison is significant, $+0.081$, $p = 0.027$, which reflects the two excluded databases). What differs dramatically is *API efficiency and precision* (Fig. A2(c)): LLM-FK required 6,026 LLM calls (original) and 6,380 (anonymous) versus only 958 prior calls + 167 verifier calls = 1,125 total calls for RefSel ($5.4\times$ fewer calls and $13\times$ fewer input tokens), while LLM-FK's pooled precision dropped to 0.17 (original) and 0.10 (anonymous) compared with 0.42–0.62 for the RefSel and GBDT pipelines.

Two post hoc stratified columns in Table A12 (*Non-Wiki* and *Unique*) clarify why absolute F1 across all 41 databases is around 0.49: 30 of the 41 databases come from WikiDBs, where "declared foreign keys" were automatically mined from Wikidata property labels pointing overwhelmingly into non-unique columns that violate the relational definition of a key (so every method scores 0.22–0.35 on WikiDBs, and Stage 1 budget truncation occurred on 11 wide WikiDBs schemas and MIMIC-IV). On the 11 conventional *Non-Wiki* schemas, RefSel+prior+veto achieves 0.897 macro F1 versus 0.821 for data-only GBDT (0.883 with veto) and 0.754 vs. 0.720 anonymously (with LLM-FK scoring 0.906 on the 9 non-Wiki schemas it could process). Similarly, restricting evaluation to the 411 declared foreign keys whose target column is actually unique in the instance (*Unique* column in Table A12), RefSel+prior+veto leads at 0.680 versus 0.628 (GBDT) and 0.646 (GBDT+veto). Finally, the empirical Stage 5 retention results in Round 2 were: at $\alpha = 0.20$, realized FDP stayed safely below target on every line (0.03–0.06 original, 0.00–0.12 anonymous; at $\alpha = 0.10$, only anonymous RefSel with $\sigma(t_e)$ slightly exceeded target at 0.12), while CTU-29 calibration kept retained sets conservative (1.6 vs. 2.0 FKs/DB for RefSel c_e vs. GBDT $\sigma(t_e)$ with names, and 0.2 vs. 1.8 without names, Holm $p = 0.03$).

TABLE A12
Second independent round (41 databases, 10 source groups): decoded-set macro F1 over all databases, over the 11 non-WikiDB databases, and over declared keys with unique targets (the last two splits are post hoc); LLM calls in the respective condition. LLM-FK could not be run on two databases (terms), which count as empty predictions here; on the 39 it ran on it scores 0.432 (original) and 0.326 (anonymous). The 167 verifier calls are shared by the RefSel and GBDT lines.

Names	Line	All	Non-Wiki	Unique	Calls
original	GBDT	0.456	0.821	0.628	0
original	RefSel	0.438	0.819	0.595	0
original	RefSel + prior	0.485	0.887	0.642	958
original	RefSel + prior + veto	**0.493**	**0.897**	**0.680**	1,125
original	GBDT + veto	0.467	0.883	0.646	167
original	GBDT + prior + veto (post hoc)	0.465	0.893	0.630	1,125
original	LLM-FK (same model)	0.411	0.741	0.526	6,026
anonymous	GBDT	**0.400**	0.720	**0.536**	0
anonymous	RefSel	0.391	**0.754**	0.526	0
anonymous	LLM-FK (same model)	0.310	0.545	0.391	6,380

I.7 RQ5 (Round 3): Large-scale registered calibration and test

Pre-registered design of Round 3 (endpoints P1–P4): Rounds 1 and 2 left three methodological limitations unresolved: (i) calibrating LTT on only $n = 24$ CTU groups forced near-zero empirical loss tolerances and small retained sets; (ii) testing on only 9 and 10 source groups left comparisons against LLM-FK statistically underpowered; and (iii) calibrating on CTU while testing on RelBench/WikiDBs introduced an intentional repository shift between calibration and test. Round 3 addresses these limitations by assembling a fresh pool of 149 databases (149 registered database-level risk units) from sources untouched by any prior experiment: 32 conventional relational schemas (from Mini-Interact [34], ScienceBenchmark [35], LDBC SNB [36], RelBench Amazon, and the advising schema of Finegan-Dollak et al. [37]) plus 117 additional WikiDBs schemas. Prior to generating any predictions, we partitioned these 149 units via a pre-registered, stratum-balanced random split (with fixed random seed) into an *calibration part of* $n = 89$ *registered units* and a *test part of* $n = 60$ *units* (13 conventional and 47 WikiDBs schemas), reusing the exact frozen models from Round 2. Four schemas with ambiguous redistribution licenses were withheld from external LLM calls, and on 14 ultra-wide WikiDBs schemas where Stage 1 hit its runtime cap, all methods shared the resulting partial candidate sets (six test schemas among them skipped the LLM prior under our pre-registered rule requiring a complete candidate file for prompt construction).

We pre-registered four confirmatory endpoints: (**P1**, *risk validity*) realized unit-mean FDP on the 60 test units at $\alpha \in \{0.10, 0.20\}$ for every retention line; (**P2**, *usefulness of retention*) correct foreign keys retained per database at $\alpha = 0.20$ for GBDT+prior+veto with verifier-weighted sigmoid confidence, with a CTU-calibrated contrast registered as secondary; the implementation scope of this retention endpoint is distinguished from the supplementary RefSel contrasts below; (**P3**, *bounded-LLM lift*) decoded macro F1 of each scorer (RefSel and GBDT) with bounded LLM prior + verifier versus the exact same scorer without LLM evidence (one-sided Wilcoxon test with Holm correction across the two scorers); and (**P4**, *comparison with LLM-FK*) decoded macro F1 of RefSel+prior+veto against LLM-FK rerun with the same backbone model on shared candidates (two-sided Wilcoxon test). Comparing data-only RefSel against data-only GBDT was pre-registered as secondary.

Results of Round 3: Table A13 and Fig. 7 report the one-time Round 3 evaluation (all 2,323 automated verification checks passed). Registered endpoints and supplementary contrasts are distinguished as follows: first, **P3 (bounded-LLM lift) is strongly confirmed for both scorers** (Fig. 7(a)): adding the bounded prior and verifier lifts RefSel's decoded macro F1 across all 60 test databases from 0.402 to 0.476 ($+0.074$, 95% CI $[0.025, 0.121]$, Holm-corrected $p < 10^{-4}$, 35 wins vs. 11 losses) and lifts GBDT from 0.370 to 0.465 ($+0.095$, CI $[0.028, 0.162]$, Holm $p < 10^{-4}$). Stratified by schema type (Table A13), on the 13 conventional databases the LLM evidence boosts macro F1 by $+0.189$ (RefSel: $0.630 \rightarrow 0.819$) and $+0.231$ (GBDT: $0.575 \rightarrow 0.806$) with 11 wins and 0 losses for both scorers, while on the 47 noisy WikiDBs schemas it adds $+0.042$ and $+0.057$. Second, **the empirical P1 criterion holds across every reported line**: in both naming conditions, the realized unit-mean FDP of every retention configuration stays strictly below α, reaching at most 0.09 at $\alpha = 0.10$ and a maximum of approximately 0.16 at $\alpha = 0.20$ (with one-sided 90% bootstrap upper bounds up to 0.202 at $\alpha = 0.20$). Third, the saved GBDT+veto retention line yields 0.58 correct foreign keys per database under either calibration set. In a supplementary cross-calibration contrast on fixed T3 predictions, verifier-weighted sigmoid confidence for RefSel retains 0.55 correct foreign keys per database with the CTU calibration set and 1.97 with the Round 3 pool ($+1.42$, 95% CI $[1.00, 1.85]$), at a realized FDP of 0.14. This does not establish the preregistered GBDT+prior+veto retention endpoint: the saved retention report thresholds GBDT edges decoded without the prior, while the GBDT+prior variants are separately evaluated for decoded F1. Fourth, for **P4 (comparison with LLM-FK)**, on the 58 license-permitting test schemas, RefSel+prior+veto has a higher whole-database macro-F1 point estimate than LLM-FK (0.467 vs. 0.445, difference $+0.022$, 95% CI $[-0.012, 0.063]$, two-sided $p = 0.19$), but the difference is not statistically significant; it issues only $1{,}769$ LLM calls versus $13{,}332$ for LLM-FK ($7.5\times$ fewer calls and $< 1/20$ of the input tokens); when restricted to declared foreign keys whose target column is unique in the data, RefSel+prior+veto significantly outperforms LLM-FK by $+0.092$ F1 ($p = 0.001$). In the anonymous condition (bottom row of Table A13), both data-only scorers significantly beat LLM-FK (0.362 and 0.352 vs. 0.270, $p = 0.003$ and 0.013, while LLM-FK still consumes $16{,}275$ calls), whereas under original names data-only scorers without LLM evidence fall below LLM-FK (-0.055 and -0.091), confirming that lexical world knowledge is essential when column names are informative. Finally, on the secondary data-only comparison, RefSel edges out GBDT by $+0.032$ with names (0.402 vs. 0.370, CI $[-0.013, 0.079]$, $p = 0.21$, 29 wins vs. 21 losses; $+0.010$ anonymous), and once both scorers receive the bounded LLM evidence they converge (0.476 vs. 0.465, $p = 0.92$). Note that across all methods, absolute whole-database F1 on the 47 WikiDBs test schemas is capped near 0.38 because only 4% of the $8{,}385$ label-mined WikiDBs keys point to unique target columns (candidate recall is 54% on unique-target WikiDBs keys and 100% on the 13 conventional schemas).

TABLE A13
Third independent round (test part: 60 databases, each its own unit; thresholds calibrated on 89 units of the same pool). Decoded macro F1 with original names over all databases and per stratum, and LLM calls. LLM-FK counts the two databases it may not process as empty; on the 58 it ran on it scores 0.445 and RefSel with prior and veto 0.467.

Line	All (60)	Conv. (13)	WikiDBs (47)	Calls
GBDT	0.370	0.575	0.313	0
RefSel	0.402	0.630	0.339	0
GBDT + prior + veto	0.465	0.806	0.370	1,769
RefSel + prior + veto	**0.476**	**0.819**	**0.381**	1,769
LLM-FK (same model)	0.430	0.688	0.359	13,332
Anonymous: GBDT 0.352, RefSel 0.362, LLM-FK 0.270 (16,275 calls)				

I.8 RQ4: Generalization, use, and cost

External literature benchmarks and comparison with LLM-FK: Table A5 in Section F reports per-database accuracy across the seven classic external benchmarks, while Fig. 8 and Fig. 9 compare RefSel directly against LLM-FK [8] on its five benchmark schemas. Across all seven external benchmarks without an LLM, RefSel achieves 0.802 macro F1 (original names) and 0.703 (anonymous, rule-selected; 0.721 full), outperforming the GBDT baseline (0.767 and 0.703), Fast-FK (0.664 and 0.532), and MC-FK (0.478). Comparing against our same-model rerun of LLM-FK across the five schemas of Fig. 8 reveals a sharp dichotomy between the two naming regimes: in the *anonymous condition* (right panel of Fig. 8), data-only RefSel beats LLM-FK by $+0.137$ macro F1 (0.711 vs. 0.574), with the widest gulf on the 370-table MusicBrainz schema where LLM-FK issues $9{,}790$ LLM calls only to reach 0.162 F1 (versus 0.428 whole-database and 0.575 evaluable-key F1 for RefSel with zero LLM calls). In the *original naming condition* (left panel of Fig. 8), LLM-FK leverages domain terminology to reach 0.877 whole-database F1 on our data versions versus 0.790 for data-only RefSel (note that the hatched bars in Fig. 8, showing 0.93–1.00, reflect LLM-FK's originally published metrics evaluated strictly inside its own profiler's candidate denominator on its own dataset versions). Activating our selective bounded LLM prior ($\mu = 5$, Section 3.6) closes this named gap almost completely—raising RefSel from 0.790 to 0.872 macro F1 across the five benchmarks while requiring only 846 total LLM calls compared with $13{,}505$ calls for LLM-FK ($16\times$ fewer calls; Fig. 9 and Table A6 in Section F). On the internal 40-database validation subset where (μ, B) were tuned, the selective prior raised F1 from 0.820 to 0.850; across all seven external benchmarks, it improves or preserves F1 on six schemas and hurts only on AdventureWorks DW ($0.755 \rightarrow 0.500$), where the LLM repeatedly favors a readable calendar-date column over an integer surrogate date key.

Sensitivity to backbone LLM generation and size: Because agentic pipelines like LLM-FK rely on the LLM to judge every candidate pair, their accuracy may fluctuate across model versions. To test model sensitivity, we re-executed both approaches replacing our default backbone (DeepSeek V4.1 Flash, `deepseek-flash`) with a $4\times$ more expensive, larger model from the preceding generation (DeepSeek V4 Pro, `deepseek-v4-pro`) [38] across both LLM-FK's five benchmarks and the nine license-permitting conventional schemas from Round 2 (Fig. 10 and Table A7 in Section F). As shown by the short horizontal arrows in Fig. 10, RefSel+LLM and GBDT+LLM are nearly invariant to swapping Flash for V4 Pro (shifting only from $0.892 \rightarrow 0.880$ and $0.890 \rightarrow 0.875$ on the five benchmarks, and from $0.899 \rightarrow 0.893$ on the nine independent schemas), because strong data logits anchor the Stage 3 decoder against LLM Perturbations (Remark 1). By contrast, LLM-FK drops by nearly five points with names ($0.877 \rightarrow 0.828$) and ten points without names ($0.574 \rightarrow 0.477$, light arrow in Fig. 10) when run with V4 Pro, which rejects more valid references (e.g., AdventureWorks recall drops from 0.93 to 0.83). Across both backbones, bounded LLM evidence matches LLM-FK's accuracy at roughly $1/13$ of the API calls on the five benchmarks ($1{,}045$ vs. $13{,}484$ with V4 Pro) and $1/4$ on the nine independent schemas, while exhibiting substantially greater stability across model choices.

Pre-registered public and enterprise ERP schemas: Table A8 in Section F details our one-time blind evaluation on the ten pre-registered databases. On the eight schemas that completed within the registered runtime and memory caps under original names, the top supervised methods finish within two points of each other: macro F1 is 0.61 for RefSel, 0.60 for GBDT, and 0.59 for both Rostin-style and Fast-FK (against GBDT, RefSel wins on 2 schemas, loses on 1, and ties on 5). Without names, rule-selected RefSel and GBDT coincide at 0.48 across the nine completed schemas (and beat Fast-FK 0.61 vs. 0.55 on the seven schemas Fast-FK completed), while pre-committed RefSel (full) reaches 0.51, matching or improving over rule-selected on seven of the nine schemas (e.g., jumping from 0.53 to 0.73 on Oracle Sales History and from 0.20 to 0.27 on Apache OFBiz).

Breaking the pre-registered results down by domain reveals both where instance-based discovery succeeds and where it hits a fundamental wall: (1) on the *five conventional public databases*, RefSel achieves 0.76 macro F1 (original) and 0.71 (anonymous); (2) on *SAP SALT* (4.7M rows), it recovers 100% (1.000 F1) of the two foreign keys whose target columns are unique in the released extract and 0% of the four organizational keys pointing to non-unique columns; (3) on *Apache OFBiz* (857 tables), evaluable-key F1 reaches 0.786 (original) and 0.393 (anonymous); but (4) on *Dolibarr* (440 tables) and anonymous *iDempiere* (912 tables), F1 collapses below 0.12. An inspection of Dolibarr explains why: in open-source ERP demonstration installs, most of the hundreds of tables contain only 1 to 3 seed rows; consequently, nearly every column of those tiny tables is trivially a UCC, and any low-cardinality column elsewhere in the database is perfectly included in them. Specifically, out of Dolibarr's 659 false-positive predictions, 557 originate from a constant ($k_e = 1$) or binary ($k_e = 2$) source column, 556 point into target tables containing at most three rows, and 114 originate from the multi-tenant constant column `entity`$= 1$. This represents the pathological limit of coincidental inclusion (Section 3.2), which no instance-based profiler can disentangle without sufficient row population. Finally, two runs exceeded our pre-registered resource caps (DNF in Table A8): Stage 1 semi-join profiling on the unsubsetted 380-table MusicBrainz dump exceeded the 4 GB memory cap, and named profiling on the 912-table iDempiere schema hit the 2-hour wall-clock limit.

CTU-14 confirmation set: To evaluate a second pre-registered cohort free of sparse ERP seed tables, we pre-

registered 14 additional databases from the CTU repository [28] (disjoint in origin from all internal, external, registered, and Round-1/2/3 calibration sets) and evaluated them once under the cross-validated configurations (Table A9 in Section F). Under original names, RefSel achieves 0.765 macro F1, trailing the GBDT baseline (0.810, difference -0.045, 95% CI $[-0.104, 0.005]$, 3 wins vs. 5 losses, Wilcoxon $p = 0.31$; the gap stems from three small schemas: PTE, Telstra, and NBA) while remaining well ahead of Fast-FK (0.723) and MC-FK (0.689). Under anonymous names, rule-selected RefSel and GBDT coincide at 0.700, outperforming MC-FK (0.689) and Fast-FK (0.654). At $\alpha = 0.20$, realized FDP on CTU-14 is 0.27 (original) and 0.19 (anonymous). Together with Round 2, this confirms that while RefSel consistently outperforms prior published baselines on new schemas, the data-only F1 margin of significance+stacking over the base GBDT context classifier does not generalize beyond BIRD/Spider.

Downstream text-to-SQL utility: To test whether discovered foreign keys improve downstream query generation, we evaluated DeepSeek V4.1 Flash on the 1,070 BIRD development questions whose gold SQL requires at least one table join (detailed in Section E), comparing schema prompts augmented with: (i) ground-truth declared foreign keys (0.550 execution accuracy), (ii) no foreign keys (0.539), (iii) all foreign keys decoded by RefSel (0.545), and (iv) only the high-confidence subset retained by RefSel at $\alpha = 0.20$ (0.546). While RefSel recovers more than half of the execution-accuracy gap between no keys and ground-truth keys—and on the 443 hardest join questions pairing differently named columns, ground-truth keys add $+0.018$ and RefSel keys add $+0.016$ while cutting missing-join errors from 209 down to 193 and 200—all overall paired differences have 95% confidence intervals straddling zero (declared minus none: $+0.011$, CI $[-0.003, 0.025]$), because BIRD's compact 3–13 table schemas with clean column names allow modern LLMs to infer most joins lexically.

Runtime scaling and monetary cost: Fig. 11 plots the wall-clock runtime of each pipeline stage against schema width on log–log axes. Stage 3 CP-SAT decoding requires only $\sim$ 1 ms per database and Stage 2 GBDT scoring completes in well under 1 s; total runtime is dominated by Stage 1 SQL semi-join candidate generation. On typical schemas Stage 1 finishes in under a second, but its cost scales with both column count and row volume—reaching 19 minutes on Dolibarr (5,755 columns), 90 minutes on BIRD's 129-million-row language corpus, and hitting the 2-hour cap on iDempiere and scale-factor-10 TPC-H/TPC-DS. Among baselines, MC-FK is the slowest statistical scorer (requiring 327 s across BIRD/Spider excluding candidate generation), while LLM-FK takes up to 3.8 hours on MusicBrainz. In monetary terms, querying the bounded LLM prior across all seven external benchmarks costs roughly $1 USD in total API fees at public list prices (Fig. 9).

REFERENCES

[1] W. Hoeffding, "Probability inequalities for sums of bounded random variables," *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 13–30, 1963.

[2] L. Perron and F. Didier, "CP-SAT, OR-Tools," Google, 2024.

[3] S. Bates *et al.*, "Distribution-free, risk-controlling prediction sets," *Journal of the ACM*, vol. 68, no. 6, 2021, article 43.

[4] A. N. Angelopoulos *et al.*, "Learn then test: Calibrating predictive algorithms to achieve risk control," *The Annals of Applied Statistics*, vol. 19, no. 2, pp. 1641–1662, 2025.

[5] R. Dunn, L. Wasserman, and A. Ramdas, "Distribution-free prediction sets for two-layer hierarchical models," *Journal of the American Statistical Association*, vol. 118, no. 544, pp. 2491–2502, 2023.

[6] M. Collins, "Discriminative training methods for hidden Markov models: Theory and experiments with perceptron algorithms," in *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2002, pp. 1–8.

[7] I. Tsochantaridis *et al.*, "Large margin methods for structured and interdependent output variables," *Journal of Machine Learning Research*, vol. 6, pp. 1453–1484, 2005.

[8] Z. Tang *et al.*, "LLM-FK: Multi-agent LLM reasoning for foreign key detection in large-scale complex databases," in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 33 201–33 228, extended version: arXiv:2603.07278.

[9] Transaction Processing Performance Council, "EGen 1.14.0: TPC-E data generator," 2026, source snapshot 055ce7e; accessed 2026-10-18. [Online]. Available: https://github.com/osdldbt/egen

[10] DBT-5 contributors, "Database test 5: PostgreSQL schema and integrity constraints," 2026, source snapshot f26e112; accessed 2026-10-18. [Online]. Available: https://github.com/osdldbt/dbt5

[11] M. Zhang *et al.*, "On multi-column foreign key discovery," *Proceedings of the VLDB Endowment*, vol. 3, no. 1–2, pp. 805–814, 2010.

[12] Z. Chen, V. Narasayya, and S. Chaudhuri, "Fast foreign-key detection in Microsoft SQL Server PowerPivot for Excel," *Proceedings of the VLDB Endowment*, vol. 7, no. 13, pp. 1417–1428, 2014.

[13] A. Rostin *et al.*, "A machine learning approach to foreign key discovery," in *Proc. 12th International Workshop on the Web and Databases (WebDB)*, 2009.

[14] L. Jiang and F. Naumann, "Holistic primary key and foreign key detection," *Journal of Intelligent Information Systems*, vol. 54, pp. 439–461, 2020.

[15] Y. Lin, Y. He, and S. Chaudhuri, "Auto-BI: Automatically build BI-models leveraging local join prediction and global schema graph," *Proceedings of the VLDB Endowment*, vol. 16, no. 10, pp. 2578–2590, 2023.

[16] J. Yang *et al.*, "Managing data constraints in database-backed web applications," in *Proc. 42nd ACM/IEEE International Conference on Software Engineering (ICSE)*, 2020, pp. 1098–1109.

[17] Databricks, "ADD CONSTRAINT clause, Databricks SQL language reference," accessed 2026-10-20.

[18] J.-E. Hellenberg *et al.*, "PRISMA: A privacy-preserving schema matcher using functional dependencies," in *Proc. 28th International Conference on Extending Database Technology (EDBT)*, 2025, pp. 297–309.

[19] S. Abedini *et al.*, "MaskSQL: Safeguarding privacy for LLM-based text-to-SQL via abstraction," arXiv:2509.23459, 2025.

[20] D. H. Wolpert, "Stacked generalization," *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992.

[21] Z. Kou and W. W. Cohen, "Stacked graphical models for efficient inference in Markov random fields," in *Proc. 2007 SIAM International Conference on Data Mining (SDM)*, 2007, pp. 533–538.

[22] P. Sen *et al.*, "Collective classification in network data," *AI Magazine*, vol. 29, no. 3, pp. 93–106, 2008.

[23] J. Li *et al.*, "Can LLM already serve as a database interface? a BIg bench for large-scale database grounded text-to-SQLs," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023.

[24] T. Yu *et al.*, "Spider: A large-scale human-labeled dataset for complex and cross-domain semantic parsing and text-to-SQL task," in *Proc. 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018, pp. 3911–3921.

[25] J. Peleška and G. Šír, "REDELEX: A framework for relational deep learning exploration," arXiv:2506.22199, 2025. [Online]. Available: https://arxiv.org/abs/2506.22199

[26] Stanford STAR, "CTU relational datasets (ReDeLEx port)," Dataset card, 2026, accessed 2026-10-18.

[27] M. Saerens, P. Latinne, and C. Decaestecker, "Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure," *Neural Computation*, vol. 14, no. 1, pp. 21–41, 2002.

[28] J. Motl and O. Schulte, "The CTU Prague relational learning repository," arXiv:1511.03086, 2015.

[29] J. Robinson *et al.*, "RelBench: A benchmark for deep learning on relational databases," in *Advances in Neural Information Processing Systems 37 (NeurIPS), Datasets and Benchmarks Track*, 2024.
[30] M. Wang *et al.*, "4DBInfer: A 4d benchmarking toolbox for graph-centric predictive modeling on relational DBs," *arXiv:2404.18209*, 2024.
[31] F. Lei *et al.*, "Spider 2.0: Evaluating language models on real-world enterprise text-to-SQL workflows," in *Proc. 13th International Conference on Learning Representations (ICLR)*, 2025.
[32] L. Vogel, J.-M. Bodensohn, and C. Binnig, "WikiDBs: A large-scale corpus of relational databases from Wikidata," in *Advances in Neural Information Processing Systems 37 (NeurIPS), Datasets and Benchmarks Track*, 2024.
[33] A. E. W. Johnson *et al.*, "MIMIC-IV clinical database demo (version 2.2)," PhysioNet, 2023.
[34] N. Huo *et al.*, "BIRD-INTERACT: Re-imagining text-to-SQL evaluation via lens of dynamic interactions," in *Proc. ICLR*, 2026, mini-Interact databases: https://huggingface.co/datasets/birdsql/mini-interact.
[35] Y. Zhang *et al.*, "ScienceBenchmark: A complex real-world benchmark for evaluating natural language to SQL systems," *Proc. VLDB Endow.*, vol. 17, no. 4, pp. 685–698, 2023.
[36] O. Erling *et al.*, "The LDBC social network benchmark: Interactive workload," in *Proc. ACM SIGMOD*, 2015, pp. 619–630.
[37] C. Finegan-Dollak *et al.*, "Improving text-to-SQL evaluation methodology," in *Proc. ACL*, 2018, pp. 351–360.
[38] DeepSeek, "Models & pricing," 2026, https://api-docs.deepseek.com/quick_start/pricing/, accessed 2026-10-23; `deepseek-flash` = DeepSeek-V4.1-Flash-1017, `deepseek-v4-pro` = DeepSeek-V4-Pro-0813.